

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:1/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

Software Test Plan Document

System Name: MySmartStudy: Digital Platform for Enhancing Learning Skills among First Year IPG Kampus Perempuan Melayu Melaka

Name : Akmal Hakimi Bin Abd Rashid

ID: 52213123466

Name : Mohamed Azeem Bin Sulaiman

ID: 52213123118

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:2/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

DOCUMENT APPROVAL

	Name	Date
Verified By: Mohamed Azeem Bin Sulaiman Akmal Hakimi Bin Abd Rashid	Mohamed Azeem Bin Sulaiman Akmal Hakimi Bin Abd Rashid	20/01/2026
Authenticated By: Supervisor	Azrai Bin Abdul Aziz 	22/01/2026
Approved By: Client		

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:3/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

TABLE OF CONTENTS

1.0 TEST POLICY	8
1.1 Software Testing Term Definition.....	9
1.1.1 Documentation and Traceability	10
1.2 Advantages of Testing	10
1.3 Testing Principles.....	11
1.4 Test Missions & Objectives	15
1.5 Communication Among Stakeholders	17
1.6 SDLC Identification	18
2.0 INTRODUCTION TO TEST PLAN	21
2.1 System Description	21
2.2 Assumptions.....	25
2.3 References	25
2.4 Document Overview	25
3.0 TEST STRATEGY	27
3.1 Test Objectives.....	27
3.2 Scope of Testing	28
3.3 Test Items.....	30
3.3.1 Features Tested	31
3.3.2 Test Design Techniques	32
3.4 Test Process	33
3.5 Item Pass/Fail Criteria	36
3.6 Suspension & Resumption Criteria	39
3.6.1 Suspension Criteria.....	39
3.6.2 Resumption Criteria.....	40
3.7 Entry and Exit Criteria	41
3.7.1 Entry Criteria	41

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:4/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

3.7.2 Exit Criteria	42
3.8 Risks and Contingencies	43
3.9 Test Levels (Black Box Testing, Risk-Based Testing)	44
3.9.1 Risk Analysis	45
3.9.2 Test Sequence	46
4.0 TEST CASE SPECIFICATION – MySmartStudy Mobile Application	47
4.1 Use Case Diagram for Overall Application	48
4.1.1 Use Case: Map Creation & Management	49
4.1.2 Use Case: Manage "My Maps"	51
4.1.3 Use Case: Collaborate on Map	52
4.1.4 Use Case: Manage Classes	53
4.1.5 Use Case: Create Assignment	54
4.1.6 Use Case: Review & Grade Maps	55
4.1.7 Use Case: View Analytics	56
4.1.8 Use Case: Access Learning Resources	57
4.1.9 Use Case: Track Achievements	58
4.1.10 Use Case: Manage Profile & Settings	59
4.1.11 Use Case: Share Teaching Resources	60
4.1.12 Use Case: View Student Dashboard	61
4.1.13 Use Case: View Lecturer Dashboard	62
4.1.14 Use Case: Export Thinking Map	63
4.1.15 Use Case: Login / Register	64
4.1.16 Use Case: Forgot Password	65
4.1.17 Use Case: Session Timeout	66
4.1.18 Use Case: Assignment Deadline Reminder Notification	67
4.1.19 Use Case: Achievement Badge Notification	69
4.1.20 Use Case: Concurrent Editing Conflict Handling	70
4.1.21 Use Case: High Traffic Load	72
4.1.22 Use Case: Responsive Design on Web	74

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:5/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

4.1.23 Use Case: Join Class Using Class Code.....	Error! Bookmark not defined.
4.1.24 Use Case: Invalid Login Attempt Handling.....	75
4.1.25 Use Case: Profile Image Upload and Update	76
4.1.26 Use Case: Search Task or Assignment by Keyword	77
4.1.27 Use Case: Delete Thinking Map with Confirmation	78
4.1.28 Use Case: Filter Thinking Maps or Task by Subject or Date	79
4.1.29 Use Case: Assignment Submission File Validation	80
4.1.30 Use Case: Multiple Device Login Synchronization.....	81
4.1.31 Use Case: Rename Thinking Map	82
4.1.32 Use Case: Assignment Late Submission Handling	83
4.1.33 Use Case: Lecturer Feedback Notification to Student.....	84
4.1.34 Use Case: Student View Assignment Submission History	85
4.1.35 Use Case: Lecturer View Class Submission Summary	86
4.1.36 Use Case: Analytics Filter by Student Submission Range.....	87
4.1.37 Use Case: Update User Profile	Error! Bookmark not defined.
4.1.38 Use Case: Create New Assignment.....	Error! Bookmark not defined.
4.1.39 Use Case: View Leaderboard Ranking	Error! Bookmark not defined.
4.1.40 Use Case: In-App Chat for Collaboration	89
4.1.41 Use Case: Duplicate Thinking Map	90
4.1.42 Use Case: Join Class via Access Code	91
4.1.43 Use Case: Scan Text to Node (OCR)	92
4.1.44 Use Case: Set Daily Study Goal	93
4.1.45 Use Case: Export Personal Data (GDPR).....	95
4.1.46 Use Case: Archive Completed Course	96
4.2 Test Case Specification	116
4.2.1 Test Case: Map Creation Module	116
4.2.2 Test Case: Collaboration Module	118
4.2.3 Test Case: Class Management Module	119
4.2.4 Test Case: Assignment & Grading Module	119

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:6/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

4.2.5 Test Case: Analytics & Reporting Module	121
4.2.6 Test Case: Utility Features	122
4.2.7 Test Case: Authentication Module	123
4.2.8 Test Case: User Profile Module.....	124
4.2.9 Test Case: Dashboard Management Module	125
4.2.10 Test Case: Search & Navigation Module	126
4.2.11 Test Case: Map Visual Customization Module	126
4.2.12 Test Case: Help and Support Module.....	127
4.2.13 Test Case: Session Management Module	128
4.2.14 Test Case: Multimedia Attachment Module	129
4.2.15 Test Case: Real-Time Collaboration Module.....	130
4.2.16 Test Case: History & Restoration Module	131
4.2.17 Test Case: Tagging & Organization Module.....	132
4.2.18 Test Case: Auto-Save Module	133
4.2.19 Test Case: Canvas Navigation Module	134
4.2.20 Test Case: Presentation Module	135
4.2.21 Test Case: Mobile Responsiveness Module	135
4.2.22 Test Case: Keyboard Shortcut Module	136
4.2.23 Test Case: Commenting Module	137
4.2.24 Test Case: Template Library Module.....	138
4.2.25 Test Case: Printing Module	139
4.2.26 Test Case: Spell Check Module	140
4.2.27 Test Case: Trash & Recovery Module	141
4.2.28 Test Case: Offline Mode Handling Module	142
4.2.29 Test Case: Cross-Map Linking Module	143
4.2.30 Test Case: Drag and Drop Organization Module	144
5.0 TEST ENVIRONMENT	155
5.1 Testing Sites.....	155
5.1.1 Developer / Testing Site	155

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:7/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

5.2 Facilities Required	155
5.3 Hardware and Software Preparation.....	157
5.3.1 Hardware Preparation	157
5.3.2 Software Preparation.....	157
5.4 Test Tools	158
5.5. Preparation and Training Required of the Test Team	159
6.0 TEST SCHEDULE	160
7.0 DEFINITIONS AND ABBREVIATION.....	164
7.1 Definitions.....	164
7.2 Abbreviations	165
APPENDIX A -	167

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:8/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

1.0 TEST POLICY

This section outlines the testing philosophy and policies used in MySmartStudy, a full-stack Learning Management System developed for students and lecturers at IPGM Kampus Perempuan Melayu Melaka. The testing policy emphasizes compliance with organized, systematic, and industry-standard best practices, guided by software quality principles such as ISTQB and ISO 9000. Although the project is developed in an academic setting, the testing process is designed to simulate a professional software development environment. Strong emphasis is placed on quality assurance to ensure that the platform effectively supports teaching, learning, and student adaptation through reliable digital learning tools.

Testing is a critical phase of the Software Development Life Cycle (SDLC), particularly in ensuring that MySmartStudy's multi-platform architecture functions correctly and consistently. The policy is intended to verify that the integrated architecture consisting of the Next.js web frontend, Flutter mobile frontend, FastAPI backend, Firebase Authentication, and Firebase Firestore database satisfies both functional and non-functional requirements. Testing also ensures proper synchronization and consistent data flow between the web and mobile platforms through shared API services and Firebase-based authentication.

The main objective of testing is to validate the functionality, integration, and reliability of the system's major modules, including the mind map editor and viewer, course management, assignments and quizzes, discussion forum and class chat, attendance and certificates, private messaging, notifications, planner/reminders, analytics, and AI-powered features such as the AI Study Companion, AI Timetable Analysis, AI Exam Planner, AI Study Materials, AI Plagiarism Detection, and AI Grading Assistant. Defects within the web, mobile, backend, and cloud-based components are identified early, documented carefully, and corrected through systematic testing to ensure that MySmartStudy remains dependable, secure, and effective as a digital learning platform.

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:9/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

1.1 Software Testing Term Definition

Software testing is the systematic process of evaluating the MySmartStudy platform to identify defects, verify functionality, and ensure that the system operates according to its specified requirements. Testing involves examining and executing the major components of the system, including the Next.js web application, Flutter mobile application, FastAPI backend, Firebase Authentication, Firebase Firestore database, and AI-supported modules, with the objective of detecting errors and confirming that actual results are consistent with expected results. In addition, testing ensures that shared services, cross-platform data flow, and user interactions function correctly across both web and mobile environments. This process is essential to guarantee that MySmartStudy delivers reliable, secure, and effective support for teaching and learning activities. Testing encompasses both functional such as creating maps, saving notes, class management, analytics visibility and non-functional aspects like performance, usability, security, and cross-platform responsiveness. This process confirms the overall quality and reliability of the MySmartStudy application, which is crucial for supporting students and lecturers in digital learning environments.

According to ISO-9000 principles, software testing is composed of two major components:

- **Validation** - Confirmation by inspection and by the furnishing of objective. testimonies that the given requirements have been met. In simple terms, validation makes sure that the developed system is fit to the actual requirements of the user.
- **Verification** - Confirmation by observation and provision of objective. statements that required conditions to be satisfied to a specific application have been met. implemented. This would involve looking into the fact that the system was constructed in the proper way as per the. technical specifications.

During testing, the defects will be reported and logged down during testing including the wrong map rendering, authentication failures, synchronization, and inconsistency in the UI.

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:10/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

All the defects shall be given to the development team to examine and address. After fixing, the test will be re-run to find out whether the problem is solved and then regression testing will be performed to confirm that the correction has not impacted other modules of the system.

1.1.1 Documentation and Traceability

All test results, defect logs, and validation activities will be documented thoroughly to ensure transparency and traceability throughout the Waterfall development cycle of the MySmartStudy project. This supports high-quality review processes and aligns with professional software development standards.

1.2 Advantages of Testing

Testing is essential to ensure the quality, functionality, reliability, and usability of the MySmartStudy digital learning platform. Since the system is developed as a full-stack learning management system with a Next.js web frontend, Flutter mobile frontend, FastAPI backend, Firebase Authentication, and Firestore database, comprehensive testing is necessary to ensure smooth performance and a consistent user experience across multiple devices and platforms.

The platform supports different user roles, mainly students, lecturers, and administrators, each of whom depends on core system functions such as mind map creation, course management, assignments, quizzes, planner/reminders, analytics, messaging, and AI-powered learning support. Testing helps confirm that these features work correctly both as individual modules and as part of an integrated system. Through continuous and systematic testing during development, defects can be identified early before they affect actual users. Examples of defects that testing can help detect include:

- **Mind Map Feature Failures:** the mind map editor or viewer failing to save, load, render, or export maps properly across web and mobile environments.

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:11/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

- **Data Synchronization Issues:** delays or inconsistencies in updating assignments, reminders, messages, notifications, or course progress between the web and mobile platforms.
- **Analytics Errors:** lecturer dashboards showing inaccurate engagement statistics, completion tracking, or at-risk student data.
- **Notification Failures:** in-app notifications or Firebase Cloud Messaging functions not delivering important updates such as announcements or reminders.
- **Role-Based Access Errors:** students, lecturers, or admins gaining access to pages or actions outside their authorized permissions due to authentication or routing issues.
- **AI Feature Inaccuracies:** AI-powered modules such as study planning, plagiarism detection, or grading recommendations producing incorrect, incomplete, or inconsistent outputs.

Identifying such defects before deployment helps reduce user frustration, improve system stability, strengthen security, and increase confidence in the platform's effectiveness for teaching and learning.

Testing also ensures **cross-platform consistency**, since MySmartStudy operates on both web and mobile platforms while sharing the same backend services and authentication flow. For example, when a student updates a planner task, submits an assignment, or reads a notification on the mobile app, the same information should be accurately reflected in the web platform. This consistency is important to maintain correct records, reliable analytics, and a seamless learning experience across the entire ecosystem

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:12/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

1.3 Testing Principles

The MySmartStudy project applies the seven core software testing principles outlined by ISTQB (International Software Testing Qualifications Board). These principles ensure that testing is carried out in a systematic, efficient, and practical manner for a full-stack digital learning platform that serves students, lecturers, and administrators. In MySmartStudy, these principles guide the testing of major system components such as the web frontend, mobile frontend, backend API, Firebase services, mind map features, course tools, messaging, notifications, analytics, and AI-powered modules throughout the software development process.

Principle 1: Testing shows the presence of defects

Testing helps uncover defects, but it cannot guarantee that the system is entirely error-free. For example, during testing of the Thinking Map creation module, issues may arise such as:

- Maps not saving correctly
- Wrong map type being recommended
- Real-time collaboration failing to update changes

These defects confirm that bugs exist even when other features appear to function well.

Principle 2: Exhaustive testing is impossible

It is not possible to test every input, action, device, user role, and usage scenario in MySmartStudy. Users may access the platform through different environments, create different types of learning content, submit assignments in multiple forms, interact through discussions and messaging, and use various AI-supported tools. Because of this, a **risk-based testing approach** should be used, with priority given to high-impact and high-risk modules such as

- **Firebase Authentication and role-based access control**
- **Cross-platform data consistency between web and mobile**
- **Assignments, quizzes, and grading workflows**
- **Notifications, messaging, and planner/reminders**

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:13/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

- **Analytics dashboards and completion tracking**
- **AI-powered features such as study planning, plagiarism detection, and grading recommendations**

Principle 3: Early testing

Testing should begin as early as possible in the software development cycle. In the MySmartStudy project, test cases will be designed during early modeling and requirement stages of the Waterfall methodology.

For example, while designing the collaboration feature, we can already create tests for:

- Map sharing permissions
- Real-time update accuracy
- Access rights for students vs. lecturers

Detecting issues early helps reduce rework, cost, and schedule delays.

Principle 4: Defect clustering

A majority of defects often reside in a small number of modules. For MySmartStudy, more complex and sensitive modules may include:

- Authentication and user synchronization
- Mind map creation, collaboration, and export features
- Assignments, quizzes, and gradebook calculations
- Messaging, notifications, and forum discussions
- Analytics dashboards and AI-supported modules

Because these modules involve more complex logic, frequent data updates, or integration across multiple services, they require closer attention and repeated testing.

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:14/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

Principle 5: Pesticide paradox

Running the same test cases repeatedly will eventually fail to uncover new defects. To overcome this, test cases must be regularly updated, especially when new features or enhancements are introduced. For example:

If new AI study tools, dashboard filters, or planner functions are added, new test scenarios should be created to verify correctness, usability, and performance. Updating test cases regularly helps maintain the effectiveness of the testing process:

- Filter accuracy
- Sorting behaviour
- Data loading performance

Refreshing test scenarios helps maintain defect discovery effectiveness.

Principle 6: Testing is context dependent

The testing approach must match the context, purpose, and user environment of the system. MySmartStudy is an academic digital-learning platform used by:

- **Students** (notetaking, Thinking Maps creation, collaboration)
- **Lecturers** (class monitoring, assignments, analytics)

Thus, testing must reflect this context by focusing on usability, clarity, educational workflow accuracy, collaboration, and performance in a learning environment.

Principle 7: Absence-of-errors fallacy

If a system is free of defects but does not fulfil user needs, it still fails. Even if MySmartStudy has no technical bugs, it must still:

- Recommend the correct Thinking Map types
- Support smooth note organization

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:15/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

- Provide meaningful analytics to lecturers
- Enable real-time collaboration without confusion
- Enhance learning skills as stated in the project goals
- Useful analytics and AI-supported study features

1.4 Test Missions & Objectives

The mission of testing in the **MySmartStudy** project is to ensure that the platform operates as a reliable, secure, and user-friendly digital learning system for **students, lecturers, and administrators**. Since the platform integrates a **web application, mobile application, backend API, Firebase services, and AI-supported features**, testing is necessary to confirm that each component functions correctly both independently and as part of the overall system. The purpose of testing is to ensure that the platform delivers accurate results, consistent cross-platform behaviour, proper role-based access, and a smooth user experience in supporting teaching and learning activities.

Specific Objectives of Testing:

- **Ensure role-based access functions correctly:** Verify that students, lecturers, and administrators can log in successfully and access only the features and pages authorized for their roles. This includes confirming that restricted actions such as user management, grading, course administration, and homepage editing are protected from unauthorized access.
- **Validate mind map creation, editing, sharing, and saving:** Confirm that users can create, modify, organize, save, and view mind maps correctly, and that map-related data is stored and retrieved accurately through the system. Testing should also verify collaboration, annotations, and export-related functions where applicable.

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:16/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

- **Verify assignments, quizzes, and grading workflows:** Ensure that lecturers can create assignments and quizzes, students can submit work and attempt quizzes successfully, and lecturers can review, grade, and provide feedback accurately through the platform.
- **Confirm communication and notification features:** Test whether discussions, forums, private messaging, announcements, and notifications function correctly so that users receive timely updates and can communicate without data loss or delivery failure.
- **Validate planner, attendance, certificates, and progress tracking modules:** Ensure that reminder tasks, attendance records, certificate claiming, calendar events, and course progress information are recorded and displayed correctly for the appropriate users.
- **Check analytics and reporting accuracy:** Validate that lecturer analytics, completion tracking, gradebook calculations, and related dashboards display correct and meaningful data based on user activity and course performance.
- **Ensure proper synchronization between web and mobile platforms:** Confirm that actions performed in one platform, such as updating reminders, submitting assignments, reading notifications, or sending messages, are accurately reflected in the other platform through the shared backend and database services.
- **Verify AI-powered features:** Ensure that AI modules such as the study companion, timetable analysis, exam planner, study material generation, plagiarism detection, and grading assistance respond correctly, return useful outputs, and integrate properly with the rest of the platform.
- **Confirm security and access control:** Ensure that unauthorized users cannot access protected resources, modify other users' data, or perform restricted operations, and that authentication and API access controls are functioning properly.

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:17/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

1.5 Communication Among Stakeholders

Effective communication among stakeholders is essential to ensure that the MySmartStudy platform is developed, tested, and delivered according to the expectations of users, developers, and academic supervisors. Since the system supports multiple user groups, Student and Lecturers clear and consistent communication helps align requirements, clarify testing needs, and resolve issues efficiently throughout the development process.

Stakeholders involved in this project include:

- Project developers, who develop and maintain the web frontend, mobile frontend, backend API, database integration, authentication, and AI-supported modules
- Testers, who execute test plans, record results, identify defects, and verify whether the system meets functional and non-functional requirements
- Supervisor or assessor, who provides academic guidance, reviews progress, and evaluates whether the project satisfies academic and technical expectations
- End users, such as students and lecturers, who use the platform and provide practical feedback on its usability, functionality, and reliability

Communication will take place through structured and well-defined channels to maintain clarity, traceability, and smooth collaboration during the Waterfall development phases.

Communication Activities and Channels

- **Testing reports and defect logs:** Testers should document and communicate test results and detected defects to the development team. These reports should include the module tested, test case reference, actual and expected results, defect description, and recommended corrective action. This helps maintain traceability and supports systematic defect resolution.

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:18/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

- **Developer coordination:** Developers should communicate regularly regarding implementation progress, integration issues, bug fixes, and deployment status for the web, mobile, backend, Firebase, and AI-related components of the system.
- **Supervisor consultations:** The project supervisor or assessor should review the documentation, monitor progress, provide recommendations, and confirm that the project remains aligned with its objectives and academic requirements.
- **Team communication tools:** Communication platforms such as **WhatsApp, Google Docs, Google Meet, Trello**, or similar collaboration tools may be used to discuss tasks, share documents, monitor progress, and coordinate development and testing activities. These tools help improve transparency and teamwork throughout the project.
- **User feedback during User Acceptance Testing (UAT):** Selected students and lecturers may participate in UAT to provide feedback on system usability, feature correctness, clarity of interface, and overall satisfaction. Their feedback should be recorded and communicated back to the development team so that improvements can be made before final delivery.
- **Documentation sharing:** Important project documents such as the Software Requirements Specification, test plan, test cases, defect reports, and user feedback summaries should be shared among relevant stakeholders to ensure that all decisions and changes are properly recorded.

1.6 SDLC Identification

The development of the **MySmartStudy** project follows the **Waterfall Software Development Methodology**, as stated in the system proposal. This methodology organizes software development into structured and sequential phases, namely **Requirements Analysis, System Design, Implementation, Integration and Testing, Deployment, and Maintenance**. Each

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:19/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

phase must be completed before proceeding to the next, allowing the project to be managed in a clear and systematic manner.

In the MySmartStudy project, formal testing activities are primarily carried out after the implementation phase, when the major system components have been developed. These components include the **Next.js web frontend, Flutter mobile frontend, FastAPI backend, Firebase Authentication, Firestore database, mind map features, course management, assignments and quizzes, messaging, notifications, analytics, and AI-powered modules.** Although Waterfall is sequential in nature, the academic context of the project may still allow limited feedback and refinement when issues are identified during testing, reviews, or supervisor evaluation. This means that selected aspects of the requirements or design may be revisited if necessary to improve functionality, logic, or usability.

For example, once the **mind map module** is implemented, testing can verify that maps are created, saved, loaded, shared, and exported correctly. After this module is integrated with the backend and database services, integration testing can confirm that map data, user actions, and related updates are handled correctly across the system. In the same way, other modules such as authentication, assignments, quizzes, notifications, analytics, and AI features are tested after implementation to ensure that the entire platform operates as an integrated and reliable learning management system.

This structured approach helps ensure that MySmartStudy satisfies its functional requirements, maintains system quality, and aligns with the organized workflow expected in an academic software development project.

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:20/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

Phase	Description (Adapted for MySmartStudy)
Requirements Analysis	This phase involves identifying and documenting the system requirements for MySmartStudy. Key requirements include role-based access for students, lecturers, and administrators , course management, assignments, quizzes, mind map features, messaging, notifications, planner/reminders, attendance, certificates, analytics, and AI-supported learning tools. From a testing perspective, the overall test strategy, scope, and objectives are identified during this phase.
System Design	In this phase, the overall architecture of the platform is designed. This includes designing the interaction between the web frontend, mobile frontend, backend API, Firebase Authentication , and Firestore database , as well as planning user interfaces, API structures, data flow, and access control mechanisms. Initial test scenarios and draft test cases may also be prepared based on the design specifications.
Implementation	During this phase, actual coding is carried out. The web frontend is developed using Next.js 16 with TypeScript and Tailwind CSS , the mobile application is developed using Flutter , and the backend is developed using FastAPI with Firebase integration. Major modules such as mind maps, courses, assignments, quizzes, discussions, messaging, analytics, notifications, and AI-powered features are implemented. Unit testing may also be performed during development to verify individual functions and modules.
Integration& Testing	In this phase, all developed modules are integrated and tested to ensure proper communication and correct system behaviour. Testing includes functional testing, integration testing, role-based access testing, API testing, database-related validation, cross-

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:21/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

	platform consistency testing, and defect retesting. This phase ensures that the web platform, mobile platform, backend services, and Firebase components operate correctly as a unified system.
Deployment	After the system passes integration and testing, the platform is deployed for demonstration or controlled academic use. At this stage, User Acceptance Testing (UAT) may be conducted with selected users such as students and lecturers to obtain feedback on usability, correctness, and overall satisfaction.
Maintenance	Although long-term maintenance may be outside the immediate academic scope, this phase includes documenting known limitations, suggested improvements, and possible future enhancements. Examples may include improvements to AI features, analytics, user experience, or additional learning tools for future expansion.

Table 1: SLDC identification

2.0 INTRODUCTION TO TEST PLAN

2.1 System Description

The MySmartStudy platform is a digital learning system designed to enhance the learning skills of first-year students at IPG Kampus Perempuan Melayu Melaka. It replaces traditional note-taking and manual learning strategies with a structured, interactive digital environment that promotes critical thinking, creativity, collaboration, and self-regulated learning. The system is accessible on both mobile and web platforms and supports students and lecturers through a wide range of features, including Thinking Maps, AI-powered learning support, personalized recommendations, class management, analytics, and real-time collaboration

Key Features and Functionality

- **User Roles and Access Control**

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:22/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

MySmartStudy supports three distinct roles: Students and Lecturers each with permissions suited to their responsibilities.

- **Students** can join courses, create mind maps, submit assignments, take quizzes, view grades, and use learning tools.
- **Lecturers** can manage courses, create assignments and quizzes, grade student work, monitor progress, and view analytics.
- **Admins** can manage users, edit homepage content, and review audit logs.

• Thinking Map Creation & Management

Students can create multiple types of Thinking Maps (Circle Map, Bubble Map, Tree Map, Double Bubble Map). Key capabilities include:

- Choosing templates
- Adding and editing content
- Saving and viewing mind maps
- Sharing maps with others
- Exporting maps as PNG or PDF
- Searching maps by code or title

• Recommendation Engine

The system provides AI-supported learning tools to help students study more effectively. Features include:

- AI Study Companion chat
- AI Daily Study Guide
- AI Exam Planner
- AI Timetable Analysis
- AI-generated summaries, flashcards, and practice quizzes

• Real-Time Collaboration

Students can collaborate using shared maps where peers or lecturers can view or edit content in real time.

Features include:

- Mind map collaboration

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:23/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

- Class chat and discussion forum
- Private messaging
- Announcements and notifications
- Group-related learning support

• Learning Resources & Tutorials

The platform provides learning materials and course support tools. Features include:

- Course resources and modules
- Assignment access and submission
- Quiz participation
- Peer reviews
- Course progress tracking

• Achievement & Progress Tracking

Students earn badges, track milestones, and monitor personal study progress, encouraging consistent learning behavior and self-regulation.

• Lecturer Dashboard & Class Management

Lecturers can:

- Create and manage classes
- Assign Thinking Map–based tasks
- Review, annotate, and grade student submissions
- Access analytics on student engagement and map usage
This provides data-driven teaching insights.

• Analytics & Reporting

The analytics module presents visual insights including:

- Student engagement tracking
- Completion tracking
- Active student participation

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:24/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

- At-risk student and performance insights
This helps lecturers tailor teaching strategy
- Synchronization Across Platforms

MySmartStudy is available on both web and mobile platforms.

- Data is shared across both platforms through the same backend services
- User actions such as messages, reminders, notifications, and course updates remain consistent
- This helps ensure a smooth and connected user experience across devices

System Architecture Overview

The system uses a cross-platform, cloud-backed architecture, enabling real-time updates, secure access, and scalable performance.

1. Presentation Layer

Consists of the MySmartStudy mobile application built using **Flutter and Dart** with **Material Design 3**, as well as a web application developed using **Next.js 16**, **TypeScript**, and **Tailwind CSS** for student, lecturer, and admin dashboards. These interfaces together provide responsive and user-friendly experiences for all users.

2. Application Layer

- Manages all core business logic for the system. It uses a **FastAPI-based backend** to handle server-side processing and relies on **RESTful APIs** to establish communication between the mobile app, the web platform, and the database. This layer also integrates **Firebase Authentication** to ensure secure and role-based access for all users, and supports AI-powered features through backend services.

3. Data Layer

- Supported by **Firebase Firestore**, which enables storage and retrieval of system data. This layer is responsible for storing **mind maps, user profiles, course data, assignments, quizzes, messages, notifications, attendance, certificates, and analytics information** that support various MySmartStudy functionalities.

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:25/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

2.2 Assumptions

Ref	Assumptions
1	Functional requirements for the MySmartStudy system have been reviewed, verified, and acknowledged by all stakeholders.
2	A Software Requirements Specification (SRS) document exists and contains all functional and non-functional requirements for MySmartStudy.
3	Static testing activities such as document review, walkthroughs, and requirement analysis have been completed on the SRS
4	The test environment, tools, and resources as described in Section 4 are available, configured, and ready for use.
5	The MySmartStudy web application, mobile application, backend API, Firebase services, and database will operate with stable performance during testing.
6	All identified defects from preliminary reviews or early testing will be resolved before the formal testing phases begin.
7	Any changes or updates to requirements will be communicated immediately to the testing team for proper planning and impact analysis.

Table 2:Assumptions

2.3 References

Reference Title	Description
Software Requirements Specification (SRS)	Contains detailed functional and non-functional requirements for the MySmartStudy system, serving as the primary reference for test case creation.
IEEE Standard 829-1983	Standard for Software Test Documentation, providing guidelines on the structure and content of test plans, test cases, and test reports.
ISTQB Foundation Level Syllabus	Provides globally recognized best practices on software testing principles, techniques, testing levels, and testing lifecycle.
ISO/IEC 25010:2011	International standard defining software product quality models (e.g., functionality, performance, usability, security) used to determine quality attributes for testing.

Table 3:Reference

2.4 Document Overview

This Software Test Plan (STP) for the MySmartStudy Digital Learning Platform is structured to guide and support all testing activities throughout the project lifecycle. The document

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:26/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

ensures that testing is carried out consistently, thoroughly, and in alignment with the system requirements, Waterfall development methodology, and academic standards. The STP is organized into the following sections:

- **Section 1: Test Policy**

Defines the overall testing philosophy adopted for the MySmartStudy project. This section explains key testing terms, ISTQB-based testing principles, test missions and objectives, SDLC alignment, and communication strategies among stakeholders.

- **Section 2: Introduction to Test Plan**

Provides a complete overview of the MySmartStudy system, including its key features such as mind maps, course management, assignments, quizzes, messaging, analytics, and AI-supported features, as well as its web, mobile, backend, and Firebase-based architecture. This section also outlines assumptions, references used, and the structural organization of the entire test plan.

- **Section 3: Test Strategy**

Details the testing scope for the MySmartStudy platform, including functional and non-functional testing coverage, testing levels (unit, integration, system, UAT), test design techniques, risk considerations, and criteria for test entry, exit, suspension, and resumption.

- **Section 4: Test Plan**

Describes the configuration of the test environment such as **web platform, mobile platform, backend API, Firebase services, and development tools**. It also outlines required deliverables, team roles and responsibilities, test data preparation, and the tools used to perform testing efficiently.

- **Section 5: Test Schedule**

Presents the testing milestones planned across FYP1 and FYP2. This includes timelines for requirement validation, test case development, system testing execution, UAT sessions, defect resolution cycles, and deployment preparation.

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:27/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

3.0 TEST STRATEGY

3.1 Test Objectives

The main objectives of testing for the MySmartStudy Digital Learning Platform are to:

- Verify Functional Requirements**
 Ensure all functional requirements in the SRS such as mind map features, course management, assignments, quizzes, messaging, notifications, planner, analytics, and AI-supported features are correctly implemented for all user roles (Students, Lecturers, and Admins).
- Validate System Reliability and Stability**
 Confirm that the MySmartStudy system operates reliably under normal and high usage conditions, including platform interaction across web and mobile environments.
- Ensure Security and Access Control**
 Test Firebase Authentication, role-based permissions, and protected routes to ensure only authorized users can access restricted features or sensitive data.
- Confirm Data Integrity and Synchronization**
 Verify that all changes such as mind map updates, course activities, submissions, messages, reminders, and notifications are accurately saved and reflected across the web and mobile platforms through the shared backend and database services.
- Evaluate Usability**
 Assess the user interface on both web and mobile for clarity, ease of use, responsiveness, and suitability for students, lecturers, and admins.
- Test Notification and Recommendation Features**
 Ensure system notifications such as announcements and reminders are delivered correctly, and verify that AI-supported functions such as study companion, study guide, timetable analysis, exam planner, study materials, plagiarism detection, and grading assistance produce appropriate outputs.

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:28/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

- **Validate Collaborative and Academic Features**

Confirm that communication tools, assignment and quiz workflows, grading, attendance, progress tracking, certificates, and other learning-related features function as intended.

- **Prepare for User Acceptance**

Ensure the system is stable, functional, and user-friendly to support UAT sessions with students and lecturers, validating its practical use as a digital learning platform.

3.2 Scope of Testing

The testing scope for MySmartStudy includes the following components:

- **Functional Testing**

Ensures all core features behave as expected:

- Students, Lecturers, and Admins will be tested to confirm that Firebase Authentication correctly validates login credentials and supports role-based access.
- Mind map creation, editing, saving, sharing, viewing, and exporting will be tested to ensure maps can be managed without errors.
- Assignments, quizzes, gradebook, attendance, and certificates will be tested to verify that academic workflows function correctly.
- Course management will be validated to ensure lecturers can manage courses and students can join and access learning materials successfully.
- Communication features such as class chat, forum, announcements, private messaging, and notifications will be tested to confirm that users can interact correctly.
- AI-supported features will be tested to verify that the system provides useful outputs for study support, timetable analysis, exam planning, study materials, plagiarism detection, and grading assistance.
- Analytics and progress tracking will be tested to ensure that activity data, completion information, and reports are calculated and displayed accurately.

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:29/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

• Non-Functional Testing

Includes validation of:

- **Performance testing**
 - Performance Testing ensures that the application responds efficiently on both mobile and web platforms.
 - It confirms that shared services such as messaging, notifications, and data updates remain responsive during normal and heavy usage.
 - It verifies that backend API calls and database access perform reliably under different load conditions..
- **Security testing**
 - Security Testing ensures that role-based access restrictions operate correctly.
 - It validates that **Firestore Authentication** controls access to protected pages and data.
 - It ensures that user information remains secure and is accessed only by authorized users.
- **Usability testing**
 - Usability Testing ensures that students, lecturers, and admins have a smooth and intuitive user experience.
 - It verifies that navigation flow is easy to follow and that interface elements are clear and readable.
 - It checks that the system is suitable for learning and teaching activities on both web and mobile platforms.
- **Compatibility testing**
 - Compatibility Testing ensures that the mobile app works correctly on supported mobile devices.
 - It confirms that the web application functions properly on major browsers.
 - It verifies that the user interface adapts correctly to different screen sizes and device types.
- **Integration Testing**
 - Integration Testing validates that different components of the system communicate correctly.

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:30/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

- It confirms that data is shared properly between the mobile app, web application, backend API, and Firebase services.
 - It verifies that the frontend communicates correctly with authentication and backend services for secure user sessions.
 - It ensures that the platform reads and updates information reliably across integrated components.
- **User Acceptance Testing (UAT)**
 - Conducted with actual students and lecturers, focusing on:
 - Learning and teaching effectiveness
 - Ease of use
 - Real-world alignment with academic needs
 - System reliability during course, assignment, quiz, messaging, and collaboration workflows.

3.3 Test Items

The primary items to be tested in the MySmartStudy application include all major components across the mobile app, web system, and backend services.

Mobile Application (Flutter for Android)

- The mobile app will be tested to ensure that all key functions operate correctly. This includes user login and registration, course access, assignments, quizzes, messaging, notifications, planner tasks, profile updates, attendance, certificates, and AI-supported features.
- Mobile-specific screens and user interactions will also be tested to ensure smooth navigation and correct data display.

Web Application (HTML/CSS/JS)

- The web application will be tested for lecturer, student, and admin tasks. These include course management, assignment and quiz management, gradebook, discussion tools, messaging, notifications, analytics review, mind map management, user management, and homepage editing.

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:31/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

- The web interface will also be checked for correct rendering, navigation flow, and compatibility across major browsers.

Backend Systems

- Integration tests will confirm seamless communication between the frontend systems and the FastAPI backend, Firebase Authentication, Firestore database, and AI-supported services.
- These components must function together seamlessly to support the MySmartStudy learning ecosystem.

These components must function together seamlessly to support MySmartStudy learning ecosystem.

3.3.1 Features Tested

A risk-based testing approach will prioritize high-impact features such as synchronization, authentication, and collaboration.

Key features include:

- **Secure Login & Authentication**
Verify role-based access (**Student, Lecturer, Admin**) and login reliability using **Firebase Authentication**.
- **Thinking Map Creation & Management**
Test mind map creation, template selection, editing, saving, sharing, viewing, and exporting.
- **AI-Supported Features**
Validate correct map suggestions based on user inputs.
- **Communication & Collaboration**
Ensure class chat, forum, messaging, announcements, notifications, and collaborative features work correctly across the platform.
- **Assignment Management**

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:32/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

Test lecturer assignment creation, student submissions, grading, feedback, and related workflows.

- **Analytics & Dashboards**

Verify that analytics, completion tracking, participation data, and student activity insights are accurate.

- **Quizzes & Gradebook**

Verify quiz creation, quiz attempts, result display, grade calculations, and gradebook accuracy.

- **Role-Based Access Control**

Ensure users can only access features and pages permitted for their assigned role.

3.3.2 Test Design Techniques

The following test design techniques will be used:

- **Equivalence Class Partitioning**

Divide input data into valid and invalid groups so that testing can be performed efficiently without repeating similar test cases. This technique will be applied to inputs such as login credentials, course fields, mind map titles, assignment forms, quiz data, and messaging inputs to confirm that the system accepts correct data while rejecting incorrect or incomplete entries.

- **Boundary Value Analysis**

To test the limits of input fields and numeric values in the system. This includes evaluating the maximum and minimum length of text fields such as titles, descriptions, and messages, as well as checking numerical boundaries for features such as grades, quiz scores, and planner priorities.

- **Use Case Testing**

To ensure that the system behaves correctly during actual usage. This includes scenarios such as creating a mind map, joining a course, submitting an assignment, taking a quiz, sending messages, receiving notifications, and lecturers reviewing student work.

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:33/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

- **Error Guessing**

Error Guessing will be applied based on tester experience and knowledge of commonly occurring issues. This technique will focus on areas more prone to failure, such as authentication errors, synchronization issues, incorrect analytics results, invalid role access, failed submissions, and notification delivery problems.

3.4 Test Process

Each testing stage is designed to ensure that system requirements are met, defects are identified early, and the platform functions reliably across mobile and web environments. The overall process includes planning, test design, test execution, defect management, reporting, and closure activities.

1. Test Planning

During this stage, the testing team defines the scope, objectives, resources, schedule, and responsibilities required for the testing effort. Activities include:

- Reviewing system requirements (mind maps, courses, assignments, quizzes, messaging, analytics, and AI-supported features).
- Identifying test items and features to be tested.
- Selecting test design techniques and test types.
- Preparing the high-level Test Plan document.
- Identifying required tools (emulators, Firebase console, Postman, browsers).
- Defining risks, constraints, and dependencies.

2. Test Analysis & Test Design

The purpose of this stage is to create detailed test cases and test data. Activities include:

Analysing the SRS and system design documents. Deriving test conditions for:

- Thinking Map creation and saving
- AI-supported features
- Real-time collaboration
- Role-based access
- Course, assignment, and quiz workflows
- Analytics and reporting

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:34/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

Designing test cases using:

- Equivalence Partitioning
- Boundary Value Analysis
- Use Case Testing
- Error Guessing

Preparing test data (sample student/lecturer/admin accounts, map titles, course codes, assignment data).

Output: Test Case Document & Test Data Sheet.

3. Test Environment Setup

The test environment replicates both mobile and web conditions to reflect real user scenarios. Components include:

- Mobile app (Flutter) installed on devices or emulators
- Web platform running on supported browsers
- Firebase Authentication configured with test users
- Firestore Database preloaded with sample users, courses, mind maps, and related records
- FastAPI backend and REST API endpoints configured and accessible
- Notification services enabled for testing alerts and updates.

This ensures accurate testing of synchronization, collaboration, and performance.

4. Test Execution

Testers execute test cases in the planned sequence. Activities include:

- Running unit-tested modules in integrated form
- Executing functional, non-functional, and integration test cases
- Logging actual results and comparing with expected results
- Validating real-time map updates
- Testing lecturer dashboard statistics
- Checking role-based access for Students and Lecturers
- Testing assignment, quiz, messaging, and notification workflows
- Triggering notifications and checking Firebase delivery

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:35/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

- Testing both mobile and web synchronization

Any deviation from expected results is flagged as a defect.

5. Defect Logging & Tracking

All identified defects are documented and assigned for resolution. Each defect record includes:

- Defect ID
- Module name (Mind Maps, Authentication, Courses, Assignments, Quizzes, Messaging, Analytics, AI Features, etc.)
- Severity and priority
- Steps to reproduce
- Expected vs. actual results
- System logs or screenshots

Developers fix defects, after which testers perform retesting and regression testing to ensure stability.

6. Test Reporting

Test progress and outcome are communicated to stakeholders through:

- Daily test execution reports
- Defect summary reports
- Test coverage analysis
- Risk and issue logs
- Final Test Summary Report at the end of the cycle

These reports help lecturers/supervisors monitor progress and confirm readiness for UAT.

7. Test Closure

Testing concludes when entry and exit criteria are met. Activities include:

- Ensuring all high-severity defects are resolved
- Completing regression cycles

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:36/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

- Confirming full coverage of functional and non-functional requirements
- Handing over test deliverables (test cases, defect logs, summary reports)
- Documenting lessons learned for future system improvements

A formal closure meeting is held with supervisor and development members to confirm testing completion.

3.5 Item Pass/Fail Criteria

For each test case executed in the MySmartStudy platform, the pass or fail decision is based on comparing the actual results with the expected results defined in the Software Requirements Specification (SRS), system design, and use case descriptions.

Pass Criteria

A test case will be marked PASS if all the following conditions are met for examples:

User Login:

- Students and Lecturers must log in successfully using valid credentials via Firebase Authentication.
- Incorrect credentials must be rejected.

Thinking Map Creation & Saving:

- Users can create, edit, save, and organize maps without errors.
- Saved maps appear immediately in the user's "My Maps" section.

AI-Supported Features:

- The system provides appropriate outputs for features such as study companion, study guide, timetable analysis, exam planner, study materials, plagiarism detection, and grading assistance.

Communication & Collaboration:

- Messages, discussions, announcements, and notifications are delivered and displayed correctly.
- Shared or collaborative functions behave correctly for authorized users.

Course & Assignment Workflow:

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:37/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

- Students can join courses, access materials, submit assignments, and take quizzes successfully.
- Lecturers can review, grade, and manage academic activities without errors.

Analytics Dashboard:

- Displays accurate statistics for map usage, student activity, related reports and assignments.

Synchronization:

- Any update such as messages, reminders, notifications, submissions, or profile changes on one platform is reflected correctly on the other platform.

Fail Criteria

A test case will be marked FAIL if any of the following occurs:

Incorrect or Missing Data Handling

- A mind map or record is saved but does not appear in the correct section.
- Data does not sync correctly between mobile and web.
- Analytics show incorrect values.
- Assignment, quiz, or message records do not appear correctly.

System Crashes, Freezes, or Performance Failures

- The app or web platform crashes when opening or using a key feature.
- The system becomes unresponsive or excessively slow.
- Backend, notification, or AI-supported functions fail to load properly.

Security or Access Violations

- A student is able to access lecturer-only or admin-only pages.
- Unauthorized users can edit or delete data that they do not own.
- Firebase Authentication fails to enforce role-based permission.

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:38/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

Feature Failure or Incorrect Behavior

- Communication or collaboration updates are not reflected correctly.
- Badges, attendance, certificates, or progress records do not update properly.
- Course join codes or user actions do not produce the correct result.
- Notifications are not received when expected.

Examples of Failures in MySmartStudy

1. Incorrect Map Synchronization

- A student edits a Thinking Map on the mobile app, but the lecturer dashboard still shows the old version.

2. Access Control Breach

- A student is able to access and modify assignment settings intended for lecturers only.

3. AI-Supported Feature Error

- The system gives an incorrect, incomplete, or irrelevant result for features such as study guide, timetable analysis, plagiarism detection, or grading assistance.

4. Communication or Collaboration Failure

- A message, notification, discussion update, or shared activity made by one user does not appear correctly for other users.

Defect Logging and Retesting

All failed test cases will be logged as defects using a tracking tool Jira, GitHub Issues, Excel defect log. Each defect entry will include:

- Defect ID
- Module (Mind Maps, Authentication, Courses, Assignments, Quizzes, Messaging, Analytics, AI Features)
- Severity (Critical / Major / Minor)
- Steps to reproduce

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:39/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

- Expected vs. actual behavior
- Screenshots or logs (if applicable)

Retesting

- Once developers fix the defect, the test case will be re-executed.
- The test will only be marked PASS if the issue is fully resolved and does not cause new defects.

Regression Testing

- After fixing any major feature, surrounding modules (maps, collaboration, analytics, authentication) will be retested to ensure no unintended side effects.

Defect Closure

- A defect will be marked Resolved/Closed only after successful retesting and confirmation that it meets expected system behavior.

3.6 Suspension & Resumption Criteria

Testing may need to be suspended when certain conditions prevent further execution or when continuing the test would produce invalid, unreliable, or unsafe results. Testing will only resume after corrective actions are completed and verified.

3.6.1 Suspension Criteria

Testing for the MySmartStudy platform will be suspended if any of the following occur:

1. Critical System Failure :

- The mobile or web application crashes repeatedly during core workflows such as map creation, collaboration, login.
- Major system modules fail to load like dashboard, My Maps, lecturer analytics.

2. Authentication or Access Control Failure

- Firebase Authentication becomes unavailable or blocks valid user roles from logging in.

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:40/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

- Severe role-based permission issues occur students accessing lecturer features.

3. Data Synchronization Failure

- Real-time sync between mobile and web consistently fails.
- Changes to Thinking Maps or assignments do not appear across platforms.

4. Backend or Database Outage

- Firebase Firestore becomes inaccessible.
- FastAPI backend services or required API endpoints fail.

5. Test Environment Instability

- Devices, emulators, or browsers required for cross-platform testing become unusable.
- Network instability prevents system access, synchronization, or backend communication.

6. Blocker Defects Identified

- Critical (Severity 1) defects prevent continuation of downstream test cases. For example, map editor failing to open, module not loading or lecturer dashboard not displaying class data.

3.6.2 Resumption Criteria

Testing may only resume once all conditions necessary for stable and reliable execution have been fulfilled.

- All blocking defects must be successfully resolved and confirmed through retesting.
- The affected system modules must return to a stable operational state.
- The full test environment must be available again, including devices, browsers, tools, test accounts, backend access, and network connectivity.
- Authentication, database, and API services must be functioning normally before testing continues.

Only after these conditions are met will testing continue according to the planned schedule.

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:41/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

3.7 Entry and Exit Criteria

The conditions that must be met before testing can begin and before testing can be concluded. These criteria ensure that testing is performed in a controlled, consistent, and measurable manner, helping verify that the MySmartStudy platform meets the expected quality standards.

3.7.1 Entry Criteria

Testing will begin only when all of the following conditions are satisfied:

1. All Functional Requirements Are Finalized and Approved

- The SRS and design documents for mind maps, courses, assignments, quizzes, messaging, notifications, authentication, analytics, and AI-supported features must be complete and approved.

2. Core Modules Are Implemented

- Core system modules must be fully implemented and unit-tested, such as mind map features, user authentication, course and assignment features, quizzes, messaging, notifications, analytics, profile management, AI-supported modules, and backend API functions.

3. Test Environment Is Ready

- The test environment must be fully prepared, including the availability of the mobile app build, a deployed web application, configured Firebase Authentication, Firestore database, backend API access, and prepared test accounts.

4. Test Cases and Test Data Are Prepared

- All high-priority test cases and corresponding test data must be written, reviewed, and approved before testing can proceed.

5. No Critical (Severity 1) Defects Are Outstanding

- There must be no outstanding critical defects from earlier development cycles, ensuring that the system is stable enough for full testing.

6. Tools and Resources Are Available

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:42/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

- All required tools and resources must be available, including physical devices, supported browsers, the Firebase Console, Postman, and reliable network access.

3.7.2 Exit Criteria

Testing will be considered complete when the following conditions are achieved:

1. All Planned Test Cases Have Been Executed

- Testing will be considered complete when all planned test cases have been fully executed, including functional, non-functional, and integration test cases.

2. All Critical and Major Defects Are Resolved

- Severity 1 defects, which are classified as critical, must be fixed immediately and successfully retested to confirm they no longer affect the system.
- Severity 2 defects, which are considered major, must also be fixed and retested to ensure proper functionality.
- Severity 3 defects, or minor issues, may remain unresolved only if the supervisor approves their acceptance and if they do not impact the overall functionality or user experience of the system.

3. Regression Testing Completed Successfully

- Ensures that fixes to one module collaboration do not break others analytics or map saving.

4. System Functionality Meets Acceptance Requirements

- All main features, including mind maps, courses, assignments, quizzes, messaging, notifications, analytics, synchronization, authentication, and AI-supported features are validated and working as described in the SRS.

5. Performance Is Stable Across Platforms

- The system performs reliably on both mobile and web during real-time usage sessions.

6. Test Summary Report Has Been Completed

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:43/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

The TSR must include:

- Final pass/fail outcomes
- Defect metrics
- Unresolved issues
- Recommendations for UAT or deployment

7. Supervisor or Stakeholder Approval Obtained

- Final approval must be obtained from the supervisor or project stakeholders, confirming that the system is ready for User Acceptance Testing or final presentation.

3.8 Risks and Contingencies

Several potential risks could impact the testing process. Recognizing these risks allows the team to plan contingencies:

- **Ambiguous or Changing Requirements:** If the project requirements are unclear or undergo late changes, test cases may become invalid. Contingency: Maintain close communication with the project supervisor and stakeholders. Freeze requirements before major testing activities and update test cases promptly when changes occur.
- **Limited Resources or Time Constraints:** The project schedule may leave little time for thorough testing, or the test team may be understaffed. Contingency: Prioritize testing of critical features (risk-based) and use automated test tools where possible to save time. Allocate extra time for regression tests and consider extending deadlines if necessary.
- **Unavailability of Test Data or Environment:** Required data sets or environments specific devices or browser versions may not be ready. Contingency: Create mock data or use virtualization simulators/emulators to mimic unavailable resources. Plan environment setup early and verify configurations in advance.
- **Technical Issues and Tool Reliability:** Testing tools or the system under test may behave unexpectedly test scripts failing or system crashes. Contingency: Keep alternate tools or manual testing procedures as backups. Document any tool-related issues and seek technical support.
- **Personnel Risks:** Key testers or developers may become unavailable illness, other assignments. Contingency: Cross-train team members on multiple features and maintain clear documentation so that others can pick up testing tasks if needed.

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:44/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

By planning for these risks, the team can reduce their impact. Each identified risk will have an owner assigned to monitor the situation and execute the contingency plan if the risk materializes.

3.9 Test Levels (Black Box Testing, Risk-Based Testing)

The testing strategy will incorporate multiple levels and approaches to ensure comprehensive coverage:

- **Unit/Component Testing:** Individual components or functions of single page of the quiz module will be tested by developers. This white-box testing validates that each part of the code performs correctly in isolation. Defects found at this level are fixed before integration.
- **Integration Testing:** After unit testing, related components for example, the mobile app, web application, backend API, authentication service, and database are tested together. This level ensures that data and control flow correctly between modules, verifying interfaces and data exchange. Integration tests will be partly white-box, with knowledge of the design, and partly black-box, focused on interactions.
- **System Testing Black Box Testing:** The complete MySmartStudy system is tested from an end-user perspective without regard to internal code structure. Testers will execute functional test cases covering features such as mind maps, courses, assignments, quizzes, messaging, notifications, analytics, and AI-supported functions, verifying that the system meets the requirements. Black-box testing techniques such as equivalence partitioning and boundary value analysis will be used to design test cases. Usability and UI consistency checks are also part of system testing.
- **Risk-Based Testing:** Testing effort will be prioritized based on risk assessment. Features deemed most critical to the project's objectives, for example authentication, role-based access, data synchronization, assignments, quizzes, analytics, and AI-supported modules, will be tested first and more thoroughly. This approach ensures that high-impact or high-probability-of-failure areas are covered early, allowing issues to be addressed promptly.
- **Regression Testing:** Whenever changes are made such as bug fixes or minor updates, a regression test suite will be run to ensure existing functionality still works. This will

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:45/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

typically focus on previously passed test cases related to the changed code.

Regression tests help maintain overall system stability.

- Acceptance Testing: In the final test level, representative end-users such as sample first-year students or instructors may perform testing to validate that the system fulfills its intended educational purpose. This may include user acceptance test (UAT) sessions or alpha/beta testing. The feedback will determine if any final adjustments are needed before release.

3.9.1 Risk Analysis

Likelihood is determined based on how frequently a feature is used and how often errors may be discovered. Impact is determined using the monetary scale defined in the reference: **1, 10, 50, 100, 500, 1000**.

Total Risk is calculated as: $R = L \times I$

Priority is determined based on the PDF's Priority Scale.

MySmartStudy – Risk Analysis Table (Following PDF Style)

Features	Likelihood	Impact	Total Risk (R = L×I)	Priority
Failed Login (Student/Lecturer)	9	1000	9000	5 (Highest Priority)
Failed Register (New Users Unable to Create Account)	9	1000	9000	5 (Highest Priority)
Failed Create / Edit Thinking Map	9	1000	9000	5 (Highest Priority)
Failed Real-Time Collaboration on Maps	9	1000	9000	5 (Highest Priority)
Failed Access to Class Dashboard (Lecturer)	9	1000	9000	5 (Highest Priority)
Failed Submit Assignment (Student)	8	1000	8000	4 (High Priority)

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:46/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

Failed Quiz Attempt / Quiz Submission	8	1000	8000	4 (High Priority)
Failed Check Learning Progress / Analytics	8	1000	8000	4 (High Priority)
Failed View Recommendations (Study Suggestions)	8	1000	8000	4 (High Priority)
Failed AI-Supported Features	8	1000	8000	4 (High Priority)
Failed Notification Delivery (Reminders, Deadlines)	8	1000	8000	4 (High Priority)
Failed Manage Classes (Add/Remove Students)	7	1000	7000	3 (Medium Priority)
Failed Access to Resource Library (Notes, Materials)	7	1000	7000	3 (Medium Priority)
Failed Update Profile / Settings	6	900	5400	3 (Medium Priority)
Failed Achievement / Badge Tracking	5	900	4500	3 (Medium Priority)
UI/UX Navigation Issues (Menu, Layout)	4	500	2000	1 (Lowest Priority)

Table 4: Risk Analysis - MySmartStudyApp

3.9.2 Test Sequence

The test sequence for the MySmartStudy application is determined based on the **risk priority** listed in the risk analysis table. Features with the highest risk (Priority 5) will be tested first to ensure stability of core functions before proceeding with lower-priority modules.

Feature	Test Sequence	Tester
Login	01	Azeem
Register	02	Azeem
Create / Edit Thinking Map	03	Akmal
Real-Time Collaboration	04	Akmal
Access Class Dashboard (Lecturer)	05	Azeem
Submit Assignment	06	Azeem
Messaging	07	Akmal
Check Analytics / Learning Progress	08	Akmal
View Study Recommendations	09	Azeem
Notification Delivery	10	Akmal

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:47/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

View AI-Supported Features	11	Azeem
Manage Classes (Add/Remove Students)	12	Akmal
Access Resource Library	13	Azeem
Update Profile / Settings	14	Akmal
Achievement / Badge Tracking	15	Akmal
UI/UX Navigation	16	Azeem

Table 5:Test Sequence – MySmartStudyApp Based on Risk Analysis

4.0 TEST CASE SPECIFICATION – MySmartStudy Mobile Application

This section describes the detailed test cases developed for the MySmartStudy application. Each test case includes a unique identifier, a description of the functionality being tested, pre-conditions, post-conditions, step-by-step test procedures, expected results, and space for actual results and pass/fail status.

The test cases have been designed based on the approved Software Requirements Specification (SRS) and follow the test sequence prioritization established in the Risk Analysis section.

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:48/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

4.1 Use Case Diagram for Overall Application

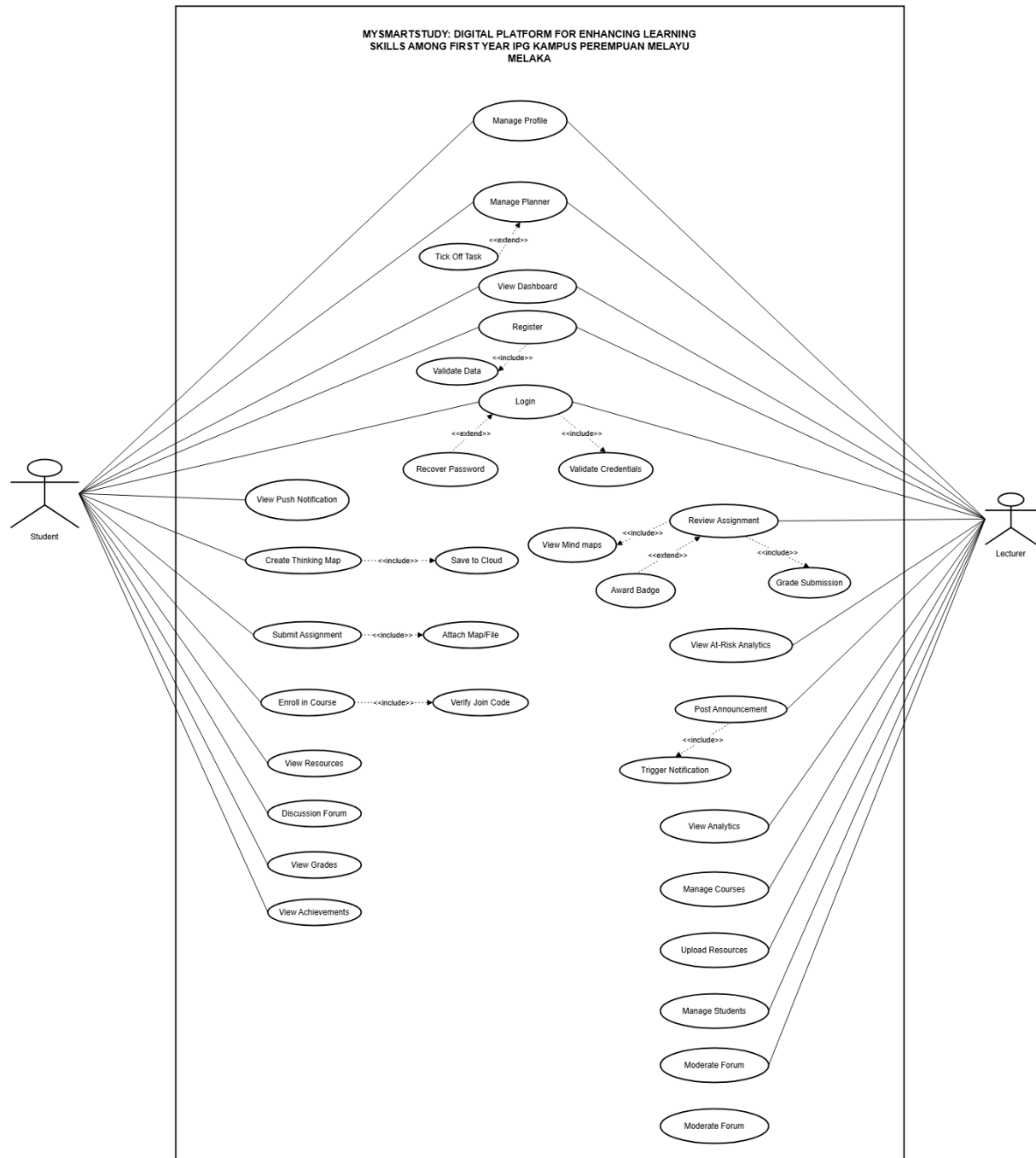


Figure 1: Use Case Diagram for overall application – MySmartStudyApp

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:49/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

4.1.1 Use Case: Map Creation & Management

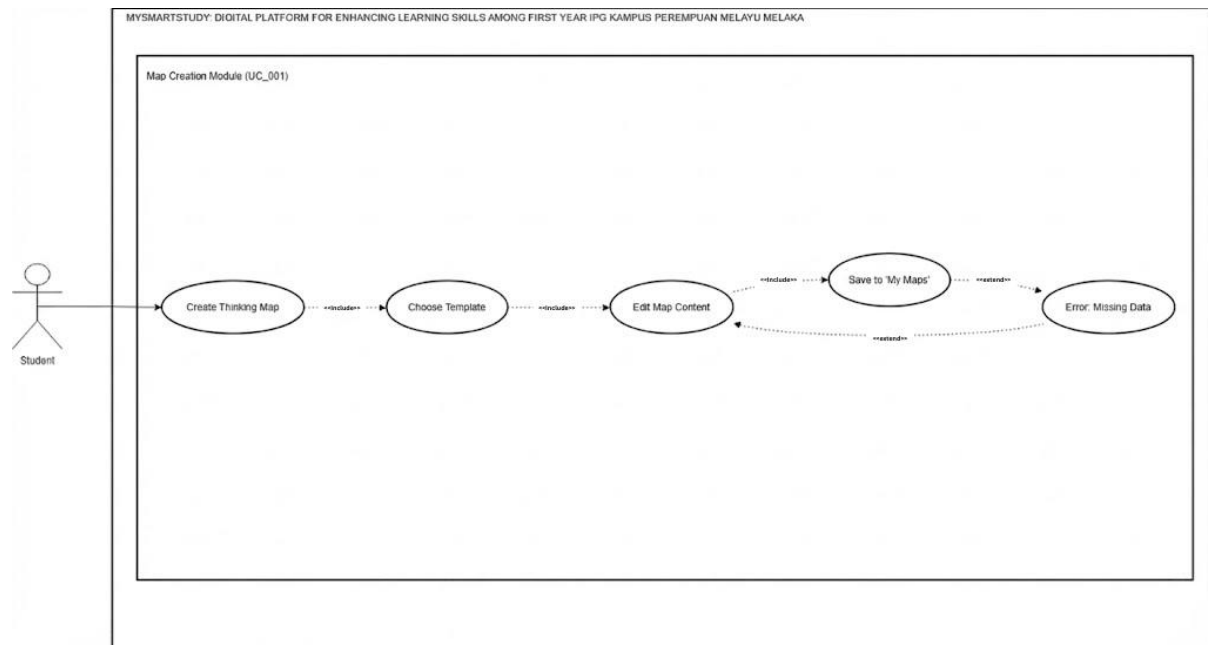


Figure 2:Use Case Diagram for UC_001– MySmartStudyApp

Use Case ID	UC_001	
Use Case Name	Create Mind Map	
Description	Allows students to create a new Thinking Map either by selecting a specific type or using the Recommendation Engine.	
Primary Actor	Student	
Include use cases	Extend: Get Map Recommendation	
Main Flow	Step	Action
	Pre-Condition	Student is logged in and on the Dashboard.
	1.1	Student selects "Create Map" from the dashboard.
	1.2	System displays options: "Choose Template" or "Get Recommendation"

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:50/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

	1.3	Student selects "Choose Template" and picks a map type (e.g., Circle Map, Bubble Map).
	1.4	System loads the selected map canvas with editing tools.
	1.5	Student enters title and content into the map nodes.
	1.6	Student clicks "Save".
	1.7	System saves the map to "My Maps" library.
	Post-Condition	A new Thinking Map is created and saved successfully.
Alternate Flow	Pre-Condition	Student is unsure which map to use.
	2.1	In step 1.2, Student selects "Get Recommendation".
	2.2	System prompts student to enter the learning task or topic description.
	2.3	System analyzes the input and suggests the most suitable Thinking Map.
	2.4	Student confirms the suggestion.
	2.5	System loads the recommended map template (Proceed to step 1.5).
	Post-Condition	Student creates a map using the AI-recommended template.
Robust Flow	Pre-Condition	Student tries to save an empty map.
	3.1	Student clicks "Save" without entering a title or content.
	3.2	System displays error message: "Map title and content cannot be empty."
	3.3	Student enters the required details.

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:51/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

	3.4	System saves the map.
	Post-Condition	The system prevents saving incomplete data.

Table 6:Use Case Statement - Map Creation (Create Thinking Map)

4.1.2 Use Case: Manage "My Maps"

Use Case ID	UC_002	
Use Case Name	Manage "My Maps"	
Description	Allows students to organize, search, filter, and export their saved Thinking Maps.	
Primary Actor	Student	
Include use cases	None	
Main Flow	Step	Action
	Pre-Condition	Student has saved maps in the system.
	1.1	Student navigates to "My Maps" library.
	1.2	System displays a list of saved maps with thumbnails and dates.
	1.3	Student selects a map to view or edit.
	1.4	System opens the map in the editor.
	Post-Condition	Student successfully accesses a previously saved map.
Alternate Flow	Pre-Condition	Student wants to organize maps by subject.
	2.1	Student selects "Create Folder".
	2.2	Student names the folder "History Notes".
	2.3	Student drags and drops specific maps into the folder.
	Post-Condition	Maps are organized into the specified folder.

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:52/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

Robust Flow	Pre-Condition	Student needs to submit the map as a PDF/Image.
	3.1	Student opens a map and clicks "Export".
	3.2	System generates a PDF or Image file of the map.
	3.3	Student downloads the file.
	Post-Condition	Map is exported to local device.

Table 7: Use Case Statement - Map Management (Manage "My Maps")

4.1.3 Use Case: Collaborate on Map

Use Case ID	UC_003	
Use Case Name	Manage collaborators	
Description	Allows a student to invite peers to view or edit a Thinking Map in real-time.	
Primary Actor	Student	
Include use cases	None	
Main Flow	Step	Action
	Pre-Condition	Student has a map open and wants to work in a group.
	1.1	Student clicks the "Share" button on the map interface.
	1.2	System prompts to enter peer's email or username.
	1.3	Student enters details and selects permission level (View or Edit).
	1.4	System sends invitation notification to the peer.
	Post-Condition	Invitation is sent, and permission is granted.

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:53/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

Alternate Flow	Pre-Condition	Peer accepts the invitation.
	2.1	Peer clicks "Join" on the notification.
	2.2	System opens the shared map for the peer.
	2.3	System enables real-time synchronization of edits.
	Post-Condition	Multiple users edit the map simultaneously.
Robust Flow	Pre-Condition	Student invites a user who does not exist.
	3.1	Student enters an invalid email address.
	3.2	System searches database and returns error: "User not found."
	3.3	Student is prompted to re-enter a valid user ID.
	Post-Condition	No invitation is sent to invalid users.

Table 8: Use Case Statement - Collaboration (Collaborate on Map)

4.1.4 Use Case: Manage Classes

Use Case ID	UC_004	
Use Case Name	Manage Courses	
Description	Enables lecturers to create courses and generate enrollment codes for students.	
Primary Actor	Lecturer	
Include use cases	None	
Main Flow	Step	Action
	Pre-Condition	Lecturer is logged in.
	1.1	Lecturer navigates to "Course Management".
	1.2	Lecturer clicks "Create New Course".
	1.3	Lecturer enters class details (Name, Subject, Semester).

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:54/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

	1.4	System creates the class and generates a unique "Class Code".
	Post-Condition	Class is created and ready for student enrollment.
Alternate Flow	Pre-Condition	Lecturer wants to remove a course.
	2.1	Lecturer selects "Delete course ".
	2.2	System requests confirmation.
	2.3	Lecturer confirms deletion.
	Post-Condition	Course and associated data are archived/deleted.
Robust Flow	Pre-Condition	Lecturer creates a course with a duplicate name.
	3.1	Lecturer enters a name that already exists for the semester.
	3.2	System warns: " Course name already exists."
	3.3	Lecturer modifies the name.
	Post-Condition	Unique class name is preserved.

Table 10: Use Case Statement - Class Management (Manage Classes)

4.1.5 Use Case: Create Assignment

Use Case ID	UC_005	
Use Case Name	Create Assignment	
Description	Allows lecturers to create tasks that require students to submit a specific type of Mind Map.	
Primary Actor	Lecturer	
Include use cases	None	
Main Flow	Step	Action
	Pre-Condition	A class exists.

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:55/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

	1.1	Lecturer selects a class and clicks "Add Assignment".
	1.2	Lecturer enters title, instructions, and due date.
	1.3	Lecturer selects the required Thinking Map type.
	1.4	Lecturer clicks "Publish".
	Post-Condition	Assignment is visible on student dashboards.
Alternate Flow	Pre-Condition	Lecturer wants to save as draft.
	2.1	Lecturer enters details but clicks "Save Draft".
	2.2	System saves assignment without publishing.
	Post-Condition	Assignment is saved but not visible to students.
Robust Flow	Pre-Condition	Lecturer sets a due date in the past.
	3.1	Lecturer selects a past date.
	3.2	System displays error: "Due date cannot be in the past."
	3.3	Lecturer corrects the date.
	Post-Condition	Valid due date is enforced.

Table 9: Use Case Statement - Assignment Creation (Create Assignment)

4.1.6 Use Case: Review & Grade Maps

Use Case ID	UC_006
Use Case Name	Review & Grade Submissions
Description	Enables lecturers to view student submissions, provide feedback, and assign grades.
Primary Actor	Lecturer
Include use cases	None

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:56/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

Main Flow	Step	Action
	Pre-Condition	Students have submitted assignments.
	1.1	Lecturer opens the "Assignments" tab and selects a submission.
	1.2	System displays the student's Mind Map.
	1.3	Lecturer adds comments/annotations and enters a grade.
	1.4	Lecturer clicks "Return".
	Post-Condition	Student receives the graded map and feedback.
Alternate Flow	Pre-Condition	Lecturer wants to filter ungraded maps.
	2.1	Lecturer selects filter "Pending Review".
	2.2	System displays only ungraded submissions.
	Post-Condition	Lecturer views relevant pending work.
Robust Flow	Pre-Condition	System fails to load student map.
	3.1	Lecturer clicks on a submission.
	3.2	System times out loading the image.
	3.3	System displays "Error loading submission. Retry?"
	Post-Condition	Lecturer can attempt to reload without data loss.

Table 10: Use Case Statement - Grading (Review & Grade Maps)

4.1.7 Use Case: View Analytics

Use Case ID	UC_007
Use Case Name	View Analytics
Description	Allows lecturers to view visual data on student engagement and map usage.
Primary Actor	Lecturer

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:57/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

Include use cases	None	
Main Flow	Step	Action
	Pre-Condition	Data exists on student activities.
	1.1	Lecturer navigates to "Analytics Dashboard".
	1.2	System retrieves data from database.
	1.3	System displays charts "Most Used Map Types".
	1.4	Lecturer filters data by class or specific student.
	Post-Condition	Lecturer gains insights into student behaviors.
Alternate Flow	Pre-Condition	Lecturer exports analytics report.
	2.1	Lecturer clicks "Export Data".
	2.2	System generates CSV/PDF report.
	Post-Condition	Analytics report is downloaded.
Robust Flow	Pre-Condition	No data available (new class).
	3.1	Lecturer opens Analytics.
	3.2	System displays message: "No data available yet."
	Post-Condition	System handles empty data states gracefully.

Table 11: Use Case Statement - Analytics (View Analytics)

4.1.8 Use Case: Access Learning Resources

Use Case ID	UC_008
Use Case Name	Access Course Resources
Description	Allows students to browse and view tutorials and example maps.
Primary Actor	Student

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:58/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

Include use cases	None	
Main Flow	Step	Action
	Pre-Condition	Student is logged in.
	1.1	Student navigates to "Learning Resources".
	1.2	Student selects a category "Map Tutorials".
	1.3	Student clicks on a specific video or article.
	1.4	System loads and displays the content.
	Post-Condition	Student successfully views learning material.
Alternate Flow	Pre-Condition	Student searches for a specific topic.
	2.1	Student enters "Bubble Map" in search bar.
	2.2	System filters resources and shows matches.
	Post-Condition	Student finds relevant specific resources.
Robust Flow	Pre-Condition	External video resource fails to load.
	3.1	Student clicks to play video.
	3.2	System detects broken link.
	3.3	System displays "Resource unavailable."
	Post-Condition	System remains stable.

Table 12: Use Case Statement - Learning Resources (Access Learning Resources)

4.1.9 Use Case: Track Achievements

Use Case ID	UC_009
Use Case Name	Track Achievements
Description	Allows students to view earned badges and study streaks.
Primary Actor	Student
Include use cases	None

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:59/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

Main Flow	Step	Action
	Pre-Condition	Student has performed activities.
	1.1	Student navigates to "Achievements" tab.
	1.2	System retrieves progress data.
	1.3	System displays badges and streak count.
	Post-Condition	Student views progress.
Alternate Flow	Pre-Condition	Student earns a new badge.
	2.1	Student completes a milestone action.
	2.2	System displays notification: "Badge Earned!"
	Post-Condition	Profile is updated with new badge.
Robust Flow	Pre-Condition	Database timeout during retrieval.
	3.1	Student opens Achievements.
	3.2	System fails to load data.
	3.3	System displays "Could not load progress."
	Post-Condition	UI loads empty state without crashing.

Table 13: Use Case Statement - Track Achievements

4.1.10 Use Case: Manage Profile & Settings

Use Case ID	UC_010	
Use Case Name	Manage Profile & Settings	
Description	Enables users to update personal information and settings.	
Primary Actor	Student, Lecturer	
Include use cases	None	
Main Flow	Step	Action
	Pre-Condition	User is logged in.

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:60/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

	1.1	User navigates to "Settings".
	1.2	User updates fields (e.g., photo, phone).
	1.3	User clicks "Save Changes".
	1.4	System updates database.
	Post-Condition	User profile is updated.
Alternate Flow	Pre-Condition	User changes password.
	2.1	User selects "Change Password".
	2.2	User enters old and new passwords.
	2.3	System validates and updates password.
	Post-Condition	Password is changed securely.
Robust Flow	Pre-Condition	User enters weak password.
	3.1	User enters password < 6 characters.
	3.2	System displays "Password too weak."
	Post-Condition	Password update is rejected.

Table 14:Use Case Statement - (Manage Profile Setting)

4.1.11 Use Case: Share Teaching Resources

Use Case ID	UC_011	
Use Case Name	Share Teaching Resources	
Description	Allows lecturers to upload materials for their classes.	
Primary Actor	Lecturer	
Include use cases	None	
Main Flow	Step	Action
	Pre-Condition	Lecturer manages a class.
	1.1	Lecturer navigates to "Resources".
	1.2	Lecturer clicks "Upload" and selects file.

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:61/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

	1.3	Lecturer adds description and clicks "Publish".
	1.4	System uploads file to cloud storage.
	Post-Condition	Resource is visible to students.
Alternate Flow	Pre-Condition	Lecturer deletes a resource.
	2.1	Lecturer selects a resource and clicks "Delete".
	2.2	System removes file from access list.
	Post-Condition	Resource is removed.
Robust Flow	Pre-Condition	Unsupported file type upload.
	3.1	Lecturer uploads .exe file.
	3.2	System rejects file: "Invalid format."
	Post-Condition	Upload is blocked.

Table 15:Use Case Statement - Sharing Teacher Resources

4.1.12 Use Case: View Student Dashboard

Use Case ID	UC_012	
Use Case Name	View Student Dashboard	
Description	Allows the student to view a summary of recent activities and progress.	
Primary Actor	Student	
Include use cases	None	
Main Flow	Step	Action
	Pre-Condition	Student logs in successfully.
	1.1	System loads Student Dashboard.
	1.2	System displays "Recent Maps", "Progress", and "Quick Actions".
	1.3	Student views overview.

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:62/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

	Post-Condition	Student is informed of current status.
Alternate Flow	Pre-Condition	New account (no history).
	2.1	System loads Dashboard.
	2.2	Displays "Welcome! Create your first map."
	Post-Condition	Empty state prompts user action.
Robust Flow	Pre-Condition	Dashboard fails to load widgets.
	3.1	System experiences lag.
	3.2	System displays cached data or loading spinners.
	Post-Condition	App does not crash.

Table 16: Use Case Statement - View Student Dashboard

4.1.13 Use Case: View Lecturer Dashboard

Use Case ID	UC_013	
Use Case Name	View Lecturer Dashboard	
Description	Allows lecturer to see class overview and pending tasks.	
Primary Actor	Lecturer	
Include use cases	None	
Main Flow	Step	Action
	Pre-Condition	Lecturer logs in successfully.
	1.1	System loads Lecturer Dashboard.
	1.2	System displays "Active Classes", "Pending Grading", and "Stats".
	1.3	Lecturer views summary.
	Post-Condition	Lecturer is updated on tasks.
Alternate Flow	Pre-Condition	Lecturer clicks on "Pending Grading" widget.
	2.1	System redirects to Grading page.

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:63/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

	Post-Condition	Navigation to specific task.
Robust Flow	Pre-Condition	Stats fail to calculate.
	3.1	System displays "Stats unavailable" placeholder.
	Post-Condition	Main navigation remains usable.

Table 17:Use Case Statement - View Lecturer Dashboard

4.1.14 Use Case: Export Mind Map

Use Case ID	UC_014	
Use Case Name	Export Mind Map	
Description	Allows a student to download a map as PDF or Image.	
Primary Actor	Student	
Include use cases	None	
Main Flow	Step	Action
	Pre-Condition	Student has a map open.
	1.1	Student clicks "Export".
	1.2	System displays format options (PDF/PNG).
	1.3	Student selects format.
	1.4	System generates and downloads file.
	Post-Condition	Map is saved locally.
Alternate Flow	Pre-Condition	Student cancels export.
	2.1	Student clicks "Cancel" on format dialog.
	2.2	System closes dialog.
	Post-Condition	No file is downloaded.
Robust Flow	Pre-Condition	Server error during conversion.
	3.1	System attempts to convert map.
	3.2	System fails and displays "Export Failed."

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:64/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

	Post-Condition	User is informed to try again.
--	----------------	--------------------------------

Table 18: Use Case Statement - Export Thinking Map

4.1.15 Use Case: Login / Register

Use Case ID	UC_015	
Use Case Name	Login / Register	
Description	Allows students and lecturers to create an account or access the system using their credentials.	
Primary Actor	Student, Lecturer	
Include use cases	None	
Main Flow	Step	Action
	Pre-Condition	User has opened the mobile app or website.
	1.1	User navigates to the "Login" screen.
	1.2	User enters Email and Password.
	1.3	User clicks "Login".
	1.4	System verifies credentials with Firebase Authentication.
	1.5	System redirects user to their specific Dashboard (Student or Lecturer).
	Post-Condition	User is authenticated and session is active.
Alternate Flow	Pre-Condition	User does not have an account.
	2.1	User selects "Register".
	2.2	User enters Full Name, Email, Password, and selects Role (Student/Lecturer).
	2.3	System creates the account in the database.
	Post-Condition	Account is created, and user is logged in.
Robust Flow	Pre-Condition	User enters invalid credentials.
	3.1	User enters wrong password.

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:65/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

	3.2	System displays error: "Invalid email or password."
	Post-Condition	User remains on Login screen.

Table 19:Use Case Statement – Login Register

4.1.16 Use Case: Forgot Password

Use Case ID	UC_016	
Use Case Name	Forgot Password	
Description	Allows users to reset their password if they have forgotten it.	
Primary Actor	Student, Lecturer	
Include use cases	None	
Main Flow	Step	Action
	Pre-Condition	User is on the Login screen.
	1.1	User clicks "Forgot Password?".
	1.2	User enters their registered Email address.
	1.3	User clicks "Send Reset Link".
	1.4	System sends a password reset email via Firebase.
	Post-Condition	User receives an email to reset credentials.
Robust Flow	Pre-Condition	User enters an email not in the database.
	2.1	User enters "unknown@example.com".
	2.2	System displays message: "If this email exists, a link has been sent." (Security best practice).
	Post-Condition	System protects user privacy by not revealing account existence.

Table 20:Use Case Statement –Forgot Password

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:66/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

4.1.17 Use Case: Session Timeout

Use Case ID	UC_017	
Use Case Name	Session Timeout	
Description	The system automatically logs out inactive users after the timeout period to ensure account security.	
Primary Actor	Student, Lecturer	
Include use cases	None	
Main Flow	Step	Action
	Pre-Condition	User is logged into the MySmartStudy app or web system.
	1.1	User logs into the system using valid credentials.
	1.2	System opens the Dashboard and starts counting inactivity time.
	1.3	User remains inactive for a defined timeout period (15 minutes).
	1.4	System automatically ends the active session.
	1.5	System redirects user to the Login page.
	Post-Condition	User is logged out automatically and must re-enter credentials to continue.
Alternate Flow	Pre-Condition	User is logged in and on the Dashboard.
	2.1	User interacts with the system (scrolls, taps, clicks).
	2.2	System resets inactivity timer.
	2.3	Session remains active and uninterrupted.

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:67/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

Robust Flow	Post-Condition	System maintains the active session without forcing logout.
	Pre-Condition	User is logged in and inactive.
	3.1	User remains inactive while network disconnects.
	3.2	System temporarily holds session locally.
	3.3	Network restores and system verifies session token.
	3.4	Token is expired, system forces logout.
	3.5	System displays message: "Session expired. Please log in again."
	Post-Condition	User must log in again; system protects account security.

Table 21:Use Case Statement – Session timeout

4.1.18 Use Case: Assignment Deadline Reminder Notification

Use Case ID	UC_018	
Use Case Name	Assignment Deadline Reminder Notification	
Description	The system sends timely reminder notifications to students before an assignment deadline.	
Primary Actor	Student	
Include use cases	None	
Main Flow	Step	Action
	Pre-Condition	A published assignment exists with a valid due date.
	1.1	System checks assignment deadlines daily.
	1.2	System detects assignment due in 24 hours.

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:68/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

	1.3	System prepares reminder notification content.
	1.4	System sends push notification to the student.
	1.5	Student receives “Assignment Due Tomorrow” alert on device.
	Post-Condition	Student is notified about the upcoming deadline.
Alternate Flow	Pre-Condition	Student submitted the assignment before the reminder date.
	2.1	System checks reminder condition.
	2.2	System identifies that student submission is completed.
	2.3	System does NOT send a deadline reminder.
	Post-Condition	Student avoids unnecessary notifications.
Robust Flow	Pre-Condition	Assignment due date is near
	3.1	System attempts to send reminder.
	3.2	Notification delivery fails due to network or FCM error.
	3.3	System retries sending after a short interval.
	3.4	On repeated failure, system logs the error for review.
	Post-Condition	System remains stable and failure is recorded for troubleshooting.

Table 22: Use Case Statement –Assignment Deadline reminder

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:69/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

4.1.19 Use Case: Achievement Badge Notification

Use Case ID	UC_019	
Use Case Name	Achievement Badge Notification	
Description	The system correctly notifies students when they earn a new achievement badge based on completed milestones.	
Primary Actor	Student	
Include use cases	None	
Main Flow	Step	Action
	Pre-Condition	Student performs an action that qualifies for a badge.
	1.1	Student completes an activity that meets the badge requirement.
	1.2	System checks achievement rules and detects a new badge earned.
	1.3	System updates the student's profile with the new badge.
	1.4	System sends a push notification: "Congratulations! You earned a new badge."
	1.5	Student receives the notification on their device.
	Post-Condition	Student's profile shows the newly earned badge, and a notification has been delivered.
Alternate Flow	Pre-Condition	Student has previously earned the badge.
	2.1	Student completes an activity matching a badge requirement.
	2.2	System checks profile and finds the badge already earned.

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:70/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

	2.3	System does not create a duplicate badge or send a notification.
	Post-Condition	Badge remains unchanged and no duplicate notification is sent.
Robust Flow	Pre-Condition	Student qualifies for a badge.
	3.1	System attempts to send badge notification.
	3.2	Notification delivery fails (network error / device offline).
	3.3	System retries sending after a defined interval.
	3.4	If failure continues, system logs the error for admin review.
	Post-Condition	Badge is still earned even if notification fails; system remains stable.

Table 23: Use Case Statement – Achievement badge notification

4.1.20 Use Case: Concurrent Editing Conflict Handling

Use Case ID	UC_020	
Use Case Name	Concurrent Editing Conflict Handling	
Description	The system handles situations where two or more collaborators edit the same Thinking Map node at the same time, ensuring data consistency and preventing conflicts.	
Primary Actor	Student	
Include use cases	None	
Main Flow	Step	Action
	Pre-Condition	Two or more students are collaborating on the same Thinking Map.

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:71/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

	1.1	Student A selects a node and begins editing text.
	1.2	System locks or marks the node as “Being edited by Student A.”
	1.3	Student B attempts to edit the same node.
	1.4	System prevents Student B from editing and displays: “This section is currently being edited.”
	1.5	Student A saves changes.
	1.6	System updates the node across all collaborator devices.
	Post-Condition	The conflict is avoided, and all users see Student A’s updated version.
Alternate Flow	Pre-Condition	Both students begin typing before the lock signal reaches both devices.
	2.1	Student A types first but hasn’t saved yet.
	2.2	Student B also types into the same node simultaneously.
	2.3	Both changes reach Firestore at slightly different times.
	2.4	System applies the latest timestamp as the final version.
	2.5	Earlier edits are overwritten.
	Post-Condition	The node displays the last submitted edit, following the timestamp-based rule.
Robust Flow	Pre-Condition	One or more collaborators have unstable network or slow device performance.

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:72/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

	3.1	Student A edits a node while Student B has temporary network delay.
	3.2	System attempts to sync Student A's update to Student B's device.
	3.3	Student B, not seeing the update yet, also edits the same node.
	3.4	System detects conflicting updates arriving out of order.
	3.5	System displays conflict message: "A newer version exists. View or overwrite?"
	3.6	Student B chooses to view latest version or retry their edit.
	Post-Condition	System prevents data corruption and informs the user of version conflict.

Table 24: Use Case Statement – Editing conflict handling

4.1.21 Use Case: High Traffic Load

Use Case ID	UC_021	
Use Case Name	High Traffic Load	
Description	The system can handle many simultaneous Firestore read/write operations.	
Primary Actor	Student, Lectures	
Include use cases	None	
Main Flow	Step	Action
	Pre-Condition	All users are logged in on their devices.
	1.1	30 students simultaneously open a Thinking Map / assignment page.

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:73/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

	1.2	Students simultaneously perform an action
	1.3	Lecturer monitors the dashboard during the load.
	1.4	System processes all simultaneous requests.
	Post-Condition	The conflict is avoided, and all users see Student A's updated version.
Alternate Flow	Pre-Condition	System is experiencing partial slowdown or delayed response.
	2.1	System experiences minor delay while processing simultaneous reads/writes.
	2.2	System queues operations and processes them sequentially.
	2.3	Users still successfully complete their tasks with only minor delays.
	Post-Condition	System remains functional, only mild slowdowns occur, no data is lost.
Robust Flow	Pre-Condition	System is under extreme load beyond expected traffic levels.
	3.1	Some requests fail due to Firestore rate limits or connection timeouts.
	3.2	System shows error message: "Unable to update now. Please retry."
	3.3	Users retry the operation.
	3.4	System successfully processes retry without corruption or duplicates.
	Post-Condition	System recovers gracefully from temporary overload and prevents inconsistent records.

Table 25: Use Case Statement – High traffic load

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:74/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

4.1.22 Use Case: Responsive Design on Web

Use Case ID	UC_022	
Use Case Name	Responsive Design on Web	
Description	The MySmartStudy web application displays correctly and remains usable across different screen sizes, browser windows, and device resolutions.	
Primary Actor	Student, Lectures	
Include use cases	None	
Main Flow	Step	Action
	Pre-Condition	Web application is deployed and accessible.
	1.1	User opens the web app at full desktop resolution.
	1.2	User navigates to Dashboard and Map Editor.
	1.3	User resizes the browser window slightly.
	1.4	Navigation menu collapses or expands appropriately.
	Post-Condition	The web interface remains visually correct and fully usable on desktop screen sizes.
Alternate Flow	Pre-Condition	User changes the browser to a medium-size window (tablet resolution).
	2.1	User resizes browser to tablet resolution.
	2.2	Sidebar becomes collapsible or moves to top menu.
	2.3	Map editor scales down to fit new width.
	Post-Condition	System remains functional and readable on medium size screens.

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:75/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

Robust Flow	Pre-Condition	User adjusts browser zoom level significantly (50% or 200%).
	3.1	User shrinks window below minimum recommended width (<500 px).
	3.2	Text and buttons scale down proportionally.
	3.3	User increases zoom to 200% or decreases to 50%.
	3.4	User opens web app in split-screen mode.
	Post-Condition	Application handles extreme screen sizes without crashing or distorting layout.

Table 26: Use Case Statement – Responsive design on web

4.1.24 Use Case: Invalid Login Attempt Handling

Use Case ID	UC_024	
Use Case Name	Invalid Login Attempt Handling	
Description	Ensures the system handles incorrect login credentials securely and informs the user appropriately.	
Primary Actor	Lecturer/Student	
Include use cases	None	
Main Flow	Step	Action
	Pre-Condition	User is on login screen.
	1.1	User enters incorrect email or password.
	1.2	User clicks “Login”.
	1.3	System detects invalid credentials.
	Post-Condition	Login is denied securely.
Alternate Flow	Pre-Condition	User retries login multiple times.
	2.1	System limits login attempts.

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:76/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

	Post-Condition	Account protection enforced.
Robust Flow	Pre-Condition	Suspicious repeated login attempts detected.
	3.1	System temporarily blocks login attempts.
	Post-Condition	Security is preserved.

Table 27: Use Case Statement – Invalid Login Attempt Handling

4.1.25 Use Case: Profile Image Upload and Update

Use Case ID	UC_025	
Use Case Name	Profile Image Upload and Update	
Description	Allows users to upload or change their profile image.	
Primary Actor	Student / Lecturer	
Include use cases	None	
Main Flow	Step	Action
	Pre-Condition	User logs in successfully.
	1.1	User navigates to Profile Settings.
	1.2	User selects “Upload Image”.
	1.3	User selects a valid image file.
		System uploads and saves image.
	Post-Condition	Profile image is updated.
Alternate Flow	Pre-Condition	New account (no history).
	2.1	System loads Dashboard.
	2.2	Displays "Welcome! Create your first map."
	Post-Condition	Empty state prompts user action.
Robust Flow	Pre-Condition	Dashboard fails to load widgets.
	3.1	System experiences lag.
	3.2	System displays cached data or loading spinners.
	Post-Condition	App does not crash.

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:77/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

Table 28: Use Case Statement – Profile Image and Update

4.1.26 Use Case: Search Task or Assignment by Keyword

Use Case ID	UC_026	
Use Case Name	Search Task or Assignment by Keyword	
Description	Allows students to quickly locate specific tasks or assignments by typing keywords into a search bar.	
Primary Actor	Student	
Include use cases	None	
Main Flow	Step	Action
	Pre-Condition	Student is on the Dashboard or Assignment list page.
	1.1	Student clicks on the search bar.
	1.2	Student types a keyword (e.g., "Math", "Chapter 1").
	1.3	System filters the list in real-time as the user types.
	1.4	System displays tasks or assignments matching the keyword.
	Post-Condition	Student sees a filtered list of relevant items.
Alternate Flow	Pre-Condition	Student searches for a non-existent term.
	2.1	Student enters a keyword with no matches.
	2.2	System displays "No results found."
	Post-Condition	User is informed that no items match the query.
Robust Flow	Pre-Condition	System experiences slow network during search.
	3.1	Student types rapidly.

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:78/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

	3.2	System delays search request until typing pauses (debounce).
	3.3	System executes search once network stabilizes.
	Post-Condition	Search results eventually appear without freezing the app.

Table 29: Use Case Statement – Search task and assignment by keyword

4.1.27 Use Case: Delete Thinking Map with Confirmation

Use Case ID	UC_027	
Use Case Name	Delete Mind Map with Confirmation	
Description	Allows a student to remove an unwanted Thinking Map but requires confirmation to prevent accidental data loss.	
Primary Actor	Student	
Include use cases	None	
Main Flow	Step	Action
	Pre-Condition	Student is viewing their list of Thinking Maps.
	1.1	Student clicks the "Delete" or "Trash" icon on a map.
	1.2	System displays a confirmation dialog: "Are you sure you want to delete this map?"
	1.3	Student confirms by clicking "Yes" or "Delete".
	1.4	System removes the map from the database.
	Post-Condition	The map is permanently deleted.
Alternate Flow	Pre-Condition	Student clicks delete by mistake.
	2.1	System displays confirmation dialog.
	2.2	Student clicks "Cancel".
	Post-Condition	The map remains in the list; no data is lost.

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:79/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

Robust Flow	Pre-Condition	Network fails during deletion request.
	3.1	Student confirms deletion.
	3.2	System attempts to delete but fails due to connection.
	3.3	System displays error: "Delete failed. Please check connection."
	Post-Condition	Map is not deleted; data integrity is preserved.

Table 30: Use Case Statement – Delete Thinking map with confirmation

4.1.28 Use Case: Filter Thinking Maps or Task by Subject or Date

Use Case ID	UC_028	
Use Case Name	Filter Mind Maps or Tasks	
Description	Enables students to organize their view by filtering items based on specific subjects or creation/due dates.	
Primary Actor	Student	
Include use cases	None	
Main Flow	Step	Action
	Pre-Condition	Student has multiple maps or tasks in their list.
	1.1	Student clicks the "Filter" button.
	1.2	Student selects a criterion (e.g., "Subject: Science" or "Date: Newest First").
	1.3	System sorts and updates the list view accordingly.
	Post-Condition	User sees the content organized by the selected filter.
Alternate Flow	Pre-Condition	Student clears the filter.
	2.1	Student clicks "Clear Filters" or "Reset".
	2.2	System restores the default list view (all items).

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:80/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

	Post-Condition	Full list of items is displayed.
Robust Flow	Pre-Condition	Student filters by a range with no data.
	3.1	Student selects a date range where no tasks exist.
	3.2	System displays an empty state message.
	Post-Condition	System correctly identifies empty result set without error.

Table 31: Use Case Statement – Filter Thinking Maps

4.1.29 Use Case: Assignment Submission File Validation

Use Case ID	UC_029	
Use Case Name	Assignment Submission File Validation	
Description	Ensures that files uploaded for assignments meet the required format (PDF/Image) and size limits.	
Primary Actor	Student	
Include use cases	None	
Main Flow	Step	Action
	Pre-Condition	Student is on the assignment submission screen.
	1.1	Student clicks "Upload File".
	1.2	Student selects a file from their device.
	1.3	System checks file type (PDF/PNG/JPG) and size (<10MB).
	1.4	System uploads the file and shows "Upload Successful".
	Post-Condition	File is attached to the assignment ready for submission.

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:81/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

Alternate Flow	Pre-Condition	Student attempts to upload an unsupported format (e.g., .exe).
	2.1	Student selects an invalid file type.
	2.2	System rejects file and displays: "Invalid file format. Please upload PDF or Image."
	Post-Condition	File is not uploaded; security is maintained.
Robust Flow	Pre-Condition	File size exceeds limit.
	3.1	Student selects a very large file (>50MB).
	3.2	System detects size violation immediately.
	3.3	System displays: "File too large. Max size is 10MB."
	Post-Condition	System prevents storage overload.

Table 32: Use Case Statement – Assignment submission file validation

4.1.30 Use Case: Multiple Device Login Synchronization

Use Case ID	UC_030	
Use Case Name	Multiple Device Login Synchronization	
Description	Ensures that actions taken on one device (e.g., tablet) are immediately reflected on another logged-in device (e.g., laptop).	
Primary Actor	Student, Lecturer	
Include use cases	None	
Main Flow	Step	Action
	Pre-Condition	User is logged in on both a tablet and a laptop.
	1.1	User creates a new Thinking Map on the tablet.
	1.2	Tablet saves data to Cloud Firestore.
	1.3	Laptop listens for database updates in real-time.

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:82/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

	1.4	Laptop automatically updates the dashboard list to show the new map.
	Post-Condition	Both devices show identical data states.
Alternate Flow	Pre-Condition	User logs out from one device.
	2.1	User clicks "Logout" on tablet.
	2.2	Tablet session ends; Laptop session remains active.
	Post-Condition	Independent session management works correctly.
Robust Flow	Pre-Condition	Network delay on one device.
	3.1	User updates profile on Laptop.
	3.2	Tablet is temporarily offline.
	3.3	Tablet reconnects to internet.
	3.4	System syncs latest data to Tablet automatically.
	Post-Condition	Data consistency is restored after reconnection.

Table 33: Use Case Statement – Multiple Design login Synchronization

4.1.31 Use Case: Rename Thinking Map

Use Case ID	UC_031	
Use Case Name	Rename Mind Map	
Description	Allows a student to change the title of an existing Thinking Map for better organization.	
Primary Actor	Student	
Include use cases	None	
Main Flow	Step	Action

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:83/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

	Pre-Condition	Student is viewing a map or the map list.
	1.1	Student selects "Rename" option for a specific map.
	1.2	System prompts for new name.
	1.3	Student enters new name and saves.
	1.4	System updates the map title in the database.
	Post-Condition	Map appears with the new name.
Alternate Flow	Pre-Condition	Student enters an empty name.
	2.1	Student clears the name field and clicks save.
	2.2	System displays validation error: "Name cannot be empty."
	Post-Condition	Rename action is blocked until valid input is provided.
Robust Flow	Pre-Condition	Name contains special forbidden characters.
	3.1	Student enters invalid characters (e.g., emojis or scripts).
	3.2	System sanitizes input or rejects it.
	Post-Condition	Database integrity is protected.

Table 34: Use Case Statement – Rename Thinking Map

4.1.32 Use Case: Assignment Late Submission Handling

Use Case ID	UC_032
Use Case Name	Assignment Late Submission Handling
Description	Manages submissions made after the official deadline, marking them as "Late" for the lecturer.
Primary Actor	Student
Include use cases	None

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:84/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

Main Flow	Step	Action
	Pre-Condition	Assignment deadline has passed.
	1.1	Student attempts to submit assignment.
	1.2	System checks current time against due date.
	1.3	System accepts the file but tags it as "Late".
	1.4	System notifies student: "Submitted Late".
	Post-Condition	Assignment is submitted with a late penalty flag.
Alternate Flow	Pre-Condition	Lecturer has disabled late submissions.
	2.1	Student attempts to submit after deadline.
	2.2	System checks settings and finds "Strict Deadline" enabled.
	2.3	System rejects submission: "Deadline passed. Submission closed."
	Post-Condition	Student cannot submit work.
Robust Flow	Pre-Condition	Submission occurs exactly at deadline minute.
	3.1	Student clicks submit at 11:59:59 PM.
	3.2	System records timestamp and accepts as "On Time".
	Post-Condition	Edge case timing is handled fairly.

Table 35: Use Case Statement – Assignment Late Submission Handling

4.1.33 Use Case: Lecturer Feedback Notification to Student

Use Case ID	UC_033
Use Case Name	Lecturer Feedback Notification to Student
Description	Notifies the student immediately when a lecturer grades or adds comments to their assignment.
Primary Actor	Student, Lecturer

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:85/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

Include use cases	None	
Main Flow	Step	Action
	Pre-Condition	Lecturer completes grading an assignment.
	1.1	Lecturer saves grade/feedback for Student A.
	1.2	System triggers a push notification to Student A.
	1.3	Student receives alert: "New feedback available for [Assignment Name]".
	1.4	Student clicks alert to view details.
	Post-Condition	Student is informed of their results.
Alternate Flow	Pre-Condition	Student is currently viewing the assignment page.
	2.1	Lecturer submits feedback.
	2.2	System updates the page in real-time without push notification.
	Post-Condition	Information is updated instantly on screen.
Robust Flow	Pre-Condition	Notification service is down.
	3.1	System fails to send push notification.
	3.2	System queues the notification for retry.
	3.3	When service restores, notification is delivered.
	Post-Condition	Student eventually receives the update.

Table 36: Use Case Statement – Lecturer feedback notification to Student

4.1.34 Use Case: Student View Assignment Submission History

Use Case ID	UC_034
Use Case Name	Student View Assignment Submission History

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:86/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

Description	Allows students to look back at previous versions of their submissions or past assignments.	
Primary Actor	Student	
Include use cases	None	
Main Flow	Step	Action
	Pre-Condition	Student is on the Assignment Details page.
	1.1	Student clicks "Submission History".
	1.2	System retrieves list of uploaded files and timestamps.
	1.3	System displays history (e.g., V1 uploaded Mon, V2 uploaded Tue).
	Post-Condition	Student can verify what was sent and when.
Alternate Flow	Pre-Condition	Student has only one submission.
	2.1	System displays the single submission details directly.
	Post-Condition	History view is simplified.
Robust Flow	Pre-Condition	History data is corrupted or missing.
	3.1	System attempts to load history.
	3.2	System displays "Unable to load history" but shows current file.
	Post-Condition	Core functionality (current file) remains accessible.

Table 37: Use Case Statement – Student View Assignment Submission History

4.1.35 Use Case: Lecturer View Class Submission Summary

Use Case ID	UC_035
Use Case Name	Lecturer View Course Submission Summary

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:87/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

Description	Provides the lecturer with an overview of how many students have submitted, who is late, and who is pending.	
Primary Actor	Lecturer	
Include use cases	None	
Main Flow	Step	Action
	Pre-Condition	Lecturer selects a specific assignment.
	1.1	Lecturer navigates to "Submission Summary".
	1.2	System calculates totals: Submitted, Late, Missing.
	1.3	System displays a dashboard/chart with these metrics.
	Post-Condition	Lecturer has a clear view of class progress.
Alternate Flow	Pre-Condition	No students have submitted yet.
	2.1	System displays 0 Submitted, 100% Pending.
	Post-Condition	Correct status is reflected.
Robust Flow	Pre-Condition	Large class size (>100 students).
	3.1	System aggregates data efficiently.
	3.2	System displays summary without UI lag.
	Post-Condition	Performance remains stable with large datasets.

Table 38: Use Case Statement – Lecturer View Assignment Submission Summary

4.1.36 Use Case: Analytics Filter by Student Submission Range

Use Case ID	UC_036
Use Case Name	Analytics Filter by Student Submission Range

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:88/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

Description	Allows lecturer to analyze performance over specific time periods (e.g., "Last 30 Days").	
Primary Actor	Lecturer	
Include use cases	None	
Main Flow	Step	Action
	Pre-Condition	Lecturer is viewing Class Analytics.
	1.1	Lecturer selects date range filter (e.g., "This Semester").
	1.2	System queries database for submissions within range.
	1.3	System recalculates average grades and completion rates.
	1.4	System updates charts to reflect filtered data.
	Post-Condition	Analytics focus on the desired timeframe.
Alternate Flow	Pre-Condition	Invalid date range selected (Start Date > End Date).
	2.1	System validation catches error.
	2.2	System displays: "Start date must be before End date."
	Post-Condition	Filter is not applied; user is corrected.
Robust Flow	Pre-Condition	No data available for selected range.
	3.1	System filters data and finds zero records.
	3.2	System displays "No activity found for this period."
	Post-Condition	Empty state is handled gracefully.

Table 39: Use Case Statement – Analytic Filter by Student Submission Range

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:89/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

4.1.40 Use Case: In-App Chat for Collaboration

Use Case ID	UC_040	
Use Case Name	In-App Chat for Collaboration	
Description	Enables real-time text communication between students while they are working on a shared Thinking Map.	
Primary Actor	Student	
Include use cases	None	
Main Flow	Step	Action
	Pre-Condition	Two students are editing the same Thinking Map.
	1.1	Student A opens the "Chat" side panel.
	1.2	Student A types a message and sends.
	1.3	Student B receives the message instantly.
	1.4	System stores the chat history for this session.
	Post-Condition	Students can coordinate edits without leaving the app.
Alternate Flow	Pre-Condition	Student receives a message while chat is closed.
	2.1	Student B sends a message.
	2.2	System shows a red notification dot on the Chat icon for Student A.
	Post-Condition	Student A is aware of new messages.
Robust Flow	Pre-Condition	Message contains profanity or banned words.
	3.1	System filters message content before sending.
	3.2	System masks the word (e.g., "*****") or blocks the message.

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:90/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

	Post-Condition	Professional environment is maintained.
--	----------------	---

Table 40: Use Case Statement – In chat App Collaboration

4.1.41 Use Case: Duplicate Thinking Map

Use Case ID	UC_041	
Use Case Name	Duplicate Mind Map	
Description	Allows a student to create an exact copy of an existing map to use as a template or for a different subject without altering the original.	
Primary Actor	Student	
Include use cases	None	
Main Flow	Step	Action
	Pre-Condition	Student is on the Dashboard or Map Editor.
	1.1	Student selects "Duplicate" from the map options menu.
	1.2	System prompts for a name for the copy (defaults to "Copy of [Name]").
	1.3	Student confirms the name.
	1.4	System copies all nodes, connections, and styles to a new database entry.
	Post-Condition	A new independent copy of the map is created.
Alternate Flow	Pre-Condition	Student cancels the duplication.
	2.1	Student clicks "Cancel" in the naming dialog.
	2.2	System aborts the operation.
	Post-Condition	No new map is created.
Robust Flow	Pre-Condition	Original map contains corrupted data.
	3.1	System attempts to clone nodes.

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:91/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

	3.2	System detects missing parent nodes or broken links.
	3.3	System displays: "Cannot duplicate. Source map is corrupted."
	Post-Condition	System prevents propagation of corrupted data.

Table 41: Use Case Statement – Duplicate Thinking Maps

4.1.42 Use Case: Join Class via Access Code

Use Case ID	UC_042	
Use Case Name	Join Class via Access Code	
Description	Allows students to self-enroll in a lecturer's course using a unique alphanumeric code provided by the lecturer.	
Primary Actor	Student	
Include use cases	None	
Main Flow	Step	Action
	Pre-Condition	Student is on the "My Classes" screen.
	1.1	Student clicks "Join Class".
	1.2	Student enters the 6-digit access code (e.g., "PHY101").
	1.3	System verifies the code against active courses.
	1.4	System enrolls the student and adds the course to their dashboard.
	Post-Condition	Student has access to the class assignments and materials.
Alternate Flow	Pre-Condition	Student enters an invalid code.

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:92/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

Robust Flow	2.1	System checks code and finds no match.
	2.2	System displays: "Invalid Class Code. Please try again."
	Post-Condition	Student is not enrolled; security is maintained.
	Pre-Condition	Student is already enrolled in the class.
	3.1	System identifies user ID is already in the class list.
	3.2	System displays: "You are already a member of this class."
	Post-Condition	Duplicate enrollment is prevented.

Table 42: Use Case Statement – Join Class Via Access Code

4.1.43 Use Case: Scan Text to Node (OCR)

Use Case ID	UC_043	
Use Case Name	Scan Text to Node (OCR)	
Description	Allows students to use their device camera to scan physical textbook text and convert it directly into a Thinking Map node.	
Primary Actor	Student	
Include use cases	None	
Main Flow	Step	Action
	Pre-Condition	Student is editing a map on a mobile device.
	1.1	Student creates a new node and selects "Scan Text".
	1.2	System opens the camera view.
	1.3	Student captures an image of the textbook page.

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:93/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

	1.4	System processes image via OCR (Optical Character Recognition).
	1.5	System inserts the extracted text into the selected node.
	Post-Condition	Physical text is digitized into the study map.
Alternate Flow	Pre-Condition	Camera permission is denied.
	2.1	System requests camera access.
	2.2	Student denies permission.
	2.3	System displays: "Camera access required for scanning."
	Post-Condition	Feature is disabled until permission is granted.
Robust Flow	Pre-Condition	Text is blurry or illegible.
	3.1	System attempts to recognize text.
	3.2	Confidence score is low.
	3.3	System displays: "Text not recognized. Please retake photo."
	Post-Condition	User is prompted to provide better input.

Table 43: Use Case Statement – Scan Text to Note (OCR)

4.1.44 Use Case: Planner / Daily Study Guide

Use Case ID	UC_044	
Use Case Name	Planner / Daily Study Guide	
Description	Allows students to set a target for time spent studying or tasks completed per day to track productivity.	
Primary Actor	Student	
Include use cases	None	
Main Flow	Step	Action
	Pre-Condition	Student is on the Dashboard.

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:94/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

	1.1	Student navigates to "Study Goals".
	1.2	Student sets a goal (e.g., "Study 2 hours" or "Complete 3 Maps").
	1.3	System saves the goal for the current day.
	1.4	System tracks activity and updates progress bar.
	Post-Condition	Student has a measurable target for the day.
Alternate Flow	Pre-Condition	Student achieves the goal.
	2.1	System detects goal completion (e.g., timer hits 2 hours).
	2.2	System triggers a "Goal Reached!" celebration animation.
	2.3	System awards bonus XP (Gamification integration).
	Post-Condition	Student is rewarded for consistency.
Robust Flow	Pre-Condition	System time setting is manipulated.
	3.1	Student changes device date to trick the system.
	3.2	System validates timestamp against server time.
	3.3	System ignores invalid activity logs.
	Post-Condition	Study tracking remains accurate and cheat-proof.

Table 44: Use Case Statement – Set Daily Study Goal

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:95/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

4.1.45 Use Case: Export Personal Data (GDPR)

Use Case ID	UC_045	
Use Case Name	Export Personal Data (GDPR)	
Description	Allows users to download a copy of all their personal data, including maps and profile info, for data portability compliance.	
Primary Actor	Student, Lecturer	
Include use cases	None	
Main Flow	Step	Action
	Pre-Condition	User is in Account Settings.
	1.1	User requests "Download My Data".
	1.2	System authenticates user again for security.
	1.3	System compiles all user records into a JSON/ZIP file.
	1.4	System emails a secure download link to the user.
	Post-Condition	User receives full copy of their stored data.
Alternate Flow	Pre-Condition	User cancels request.
	2.1	User navigates away before confirming.
	Post-Condition	No data export is initiated.
Robust Flow	Pre-Condition	Database is extremely large.
	3.1	System starts compilation in background.
	3.2	System notifies: "This may take a few minutes. We will email you."
	3.3	System processes efficiently without timing out the UI.
	Post-Condition	User experience is not blocked by heavy processing.

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:96/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

Table 45: Use Case Statement – Export Personal Data GDPR

4.1.46 Use Case: Archive Completed Course

Use Case ID	UC_046	
Use Case Name	Archive Completed Course	
Description	Allows a lecturer to hide a completed course from their active dashboard without permanently deleting the data.	
Primary Actor	Lecturer	
Include use cases	None	
Main Flow	Step	Action
	Pre-Condition	Lecturer is viewing their active courses list.
	1.1	Lecturer clicks the "Settings" icon on a course card.
	1.2	Lecturer selects "Archive Class".
	1.3	System asks for confirmation.
	1.4	System moves the course to the "Archived" tab.
	Post-Condition	Course is hidden from active view but data is preserved.
Alternate Flow	Pre-Condition	Lecturer needs to restore a class.
	2.1	Lecturer navigates to "Archived Classes".
	2.2	Lecturer clicks "Restore" on a specific course.
	2.3	System moves course back to active dashboard.
	Post-Condition	Course is once again visible to students.
Robust Flow	Pre-Condition	Course has active pending assignments.
	3.1	Lecturer attempts to archive.

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:97/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

	3.2	System checks for open assignments with future due dates.
	3.3	System warns: "Class has active assignments. Archive anyway?"
	3.4	Lecturer confirms "Yes".
	Post-Condition	Course is archived, but students may still receive due date notifications.

Table 46: Use Case Statement –Achieved Completed Course

4.1.46 Use Case: Archive Completed Course

Use Case ID	UC_047	
Use Case Name	Take Quiz	
Description	Allows students to attempt quizzes provided by lecturers for their enrolled courses.	
Primary Actor	Student	
Include use cases	None	
Main Flow	Step	Action
	Pre-Condition	Student is logged in and enrolled in a course with an available quiz.
	1.1	Student navigates to the course quiz section.
	1.2	System displays the list of available quizzes.
	1.3	Student selects a quiz.
	1.4	System loads the quiz questions and instructions.
	1.5	Student answers the questions and clicks "Submit".

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:98/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

	1.6	System records the quiz attempt and calculates the result.
	Post-Condition	Quiz attempt is saved successfully and result is displayed if allowed.
Alternate Flow	Pre-Condition	Student is not ready to start the quiz immediately.
	2.1	Student opens the quiz details page.
	2.2	Student reviews the instructions and exits without starting.
	Post-Condition	Quiz remains available for future attempt if deadline has not passed.
Robust Flow	Pre-Condition	Internet connection becomes unstable during quiz attempt.
	3.1	Student answers several questions.
	3.2	System detects connection issue.
	3.3	System displays warning message and attempts to preserve entered answers.
	3.4	Student reconnects and continues the quiz.
	Post-Condition	System prevents unnecessary data loss during the quiz attempt.

Table 47: Use Case Statement –Take Quiz

4.1.46 Use Case: View Gradebook / View Grades

Use Case ID	UC_048
Use Case Name	View Gradebook / View Grades
Description	Allows students to view their grades and lecturers to view grade records for course performance monitoring.
Primary Actor	Lecturer

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:99/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

Include use cases	None	
Main Flow	Step	Action
	Pre-Condition	Pre-Condition User is logged in and grade records exist in the system.
	1.1	User navigates to the Gradebook or Grades page.
	1.2	System retrieves grade data from the database.
	1.3	System displays assignment, quiz, and overall grade information.
	Post-Condition	User successfully views the relevant grade information.
Alternate Flow	Pre-Condition	Lecturer wants to view grades for a specific course only.
	2.1	Lecturer selects a course from the course list.
	2.2	System filters and displays grade records for that course.
	Post-Condition	Lecturer views course-specific grade details.
Robust Flow	Pre-Condition	Grade data is temporarily unavailable.
	3.1	User opens the Gradebook page.
	3.2	System fails to retrieve some records.
	3.3	System displays message: "Unable to load grades at the moment."
	Post-Condition	System remains stable and informs the user properly.

Table 48: Use Case Statement –View Gradebook / View Grades

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:100/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

4.1.46 Use Case: Send Private Message

Use Case ID	UC_049	
Use Case Name	Send Private Message	
Description	Allows students and lecturers to communicate directly through private messaging.	
Primary Actor	Lecturer, Student	
Include use cases	None	
Main Flow	Step	Action
	Pre-Condition	User is logged in.
	1.1	User navigates to the Messages page.
	1.2	User selects an existing conversation or starts a new one.
	1.3	User types a message.
	1.4	User clicks "Send".
	1.5	System stores and delivers the message to the recipient.
	Post-Condition	Message is sent successfully and appears in the conversation thread.
Alternate Flow	Pre-Condition	User wants to start a conversation with a new person.
	2.1	User clicks "New Message".
	2.2	System provides user search function.
	2.3	User selects recipient and sends first message.
	Post-Condition	New conversation is created successfully.
Robust Flow	Pre-Condition	Network interruption occurs while sending a message.
	3.1	User types and sends a message.

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:101/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

	3.2	System fails to deliver due to connection issue.
	3.3	System displays error or retry option.
	Post-Condition	User is informed that the message was not sent successfully.

Table 49: Use Case Statement –Archieved Completed Course

4.1.46 Use Case: View Notifications

Use Case ID	UC_050	
Use Case Name	View Notifications	
Description	Allows users to view system notifications such as announcements, reminders, and updates.	
Primary Actor	Student, Lecturer	
Include use cases	None	
Main Flow	Step	Action
	Pre-Condition	User is logged in and notifications exist in the system.
	1.1	User navigates to the Notifications page.
	1.2	System retrieves the notification list.
	1.3	System displays unread and read notifications.
	1.4	User opens a notification to read its details.
	Post-Condition	User successfully views notification information.
Alternate Flow	Pre-Condition	User wants to mark all notifications as read.
	2.1	User clicks "Mark All as Read".
	2.2	System updates notification status.
	Post-Condition	All visible notifications are marked as read.

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:102/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

Robust Flow	Pre-Condition	Notification data cannot be retrieved temporarily.
	3.1	User opens the Notifications page.
	3.2	System fails to load the notification list.
	3.3	System displays message: "Unable to load notifications."
	3.4	Lecturer confirms "Yes".
	Post-Condition	System remains stable and user is informed of the issue.

Table 50: Use Case Statement –View Notifications

4.1.46 Use Case: View Attendance

Use Case ID	UC_046	
Use Case Name	View Attendance	
Description	Allows students to view attendance records and lecturers to review attendance information for their courses.	
Primary Actor	Student, Lecturer	
Include use cases	None	
Main Flow	Step	Action
	Pre-Condition	User is logged in and attendance data exists.
	1.1	User navigates to the Attendance page.
	1.2	System retrieves attendance records from the database.
	1.3	System displays attendance information by course or session.
	Post-Condition	User successfully views attendance records.

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:103/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

Alternate Flow	Pre-Condition	Lecturer wants to check attendance for a specific class session.
	2.1	Lecturer selects the course and session.
	2.2	System displays student attendance statuses for that session.
	Post-Condition	Lecturer views the selected attendance details.
Robust Flow	Pre-Condition	Attendance records are missing or incomplete.
	3.1	User opens the Attendance page.
	3.2	System detects unavailable records.
	3.3	System displays message: "Attendance data not available."
	Post-Condition	System handles missing data without crashing.

Table 51: Use Case Statement – View Attendance

4.1.46 Use Case: Claim Certificate

Use Case ID	UC_046	
Use Case Name	Claim Certificate	
Description	Allows students to claim a course completion certificate when eligibility requirements are met.	
Primary Actor	Student	
Include use cases	None	
Main Flow	Step	Action
	Pre-Condition	Student is logged in and has met the course completion requirements.
	1.1	Student navigates to the Certificates page.

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:104/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

	1.2	System displays available certificates eligible for claiming.
	1.3	Student selects a certificate and clicks "Claim".
	1.4	System generates and records the certificate.
	Post-Condition	Student successfully claims the certificate.
Alternate Flow	Pre-Condition	Student wants to view previously claimed certificates.
	2.1	Student opens the Certificates page.
	2.2	System displays earned certificates.
	Post-Condition	Student views certificate history.
Robust Flow	Pre-Condition	Student has not completed the course requirements.
	3.1	Student clicks "Claim" for a certificate.
	3.2	System checks eligibility status.
	3.3	System displays message: "Certificate cannot be claimed yet."
	3.4	Lecturer confirms "Yes".
	Post-Condition	System prevents invalid certificate claiming.

Table 52: Use Case Statement – Claim Certificate

4.1.46 Use Case: AI Study Companion

Use Case ID	UC_046
Use Case Name	AI Study Companion
Description	Allows students to interact with the AI Study Companion for learning support and study guidance.
Primary Actor	Student
Include use cases	None

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:105/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

Main Flow	Step	Action
	Pre-Condition	Student is logged in.
	1.1	Student opens the AI Study Companion feature.
	1.2	System displays the chat interface.
	1.3	Student enters a question or study-related request.
	1.4	System processes the input and generates a response.
	1.5	Student reads the response and continues the conversation if needed.
	Post-Condition	Student receives AI-supported study assistance.
Alternate Flow	Pre-Condition	Student wants to continue from previous conversation history.
	2.1	Student opens the AI Study Companion.
	2.2	System loads previous chat history.
	2.3	System moves course back to active dashboard.
	Post-Condition	Student continues earlier discussion with the AI.
Robust Flow	Pre-Condition	AI service is temporarily unavailable.
	3.1	Student sends a message to the AI.
	3.2	System fails to retrieve a response.
	3.3	System displays message: "AI service unavailable. Please try again later."
	Post-Condition	System remains stable and user is informed of the issue

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:106/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

Table 53: Use Case Statement – AI Study Companion

4.1.46 Use Case: Generate AI Study Materials

Use Case ID	UC_046	
Use Case Name	Generate AI Study Materials	
Description	Allows students to generate AI-supported study materials such as summaries, flashcards, and practice quizzes.	
Primary Actor	Student	
Include use cases	None	
Main Flow	Step	Action
	Pre-Condition	Student is logged in and has access to course content or study material input.
	1.1	Student navigates to the AI Study Materials feature.
	1.2	Student selects the type of material to generate (Summary, Flashcards, or Practice Quiz).
	1.3	Student provides the required content or selects a course resource.
	1.4	System processes the request and generates the selected study material.
	1.5	System displays the generated result to the student.
	Post-Condition	Student successfully receives AI-generated study material.
Alternate Flow	Pre-Condition	Student wants to generate a different type of material from the same content.

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:107/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

	2.1	Student returns to the material options.
	2.2	Student selects another output type.
	2.3	System generates the new study material.
	Post-Condition	Student receives an alternative AI-generated study resource.
Robust Flow	Pre-Condition	Submitted content is empty or insufficient.
	3.1	Student submits the request without enough input data.
	3.2	System validates the input and displays message: "Please provide sufficient content."
	Post-Condition	System prevents invalid material generation request.

Table 54: Use Case Statement –Archieved Completed Course

4.1.46 Use Case: Generate AI Daily Study Guide

Use Case ID	UC_046	
Use Case Name	Generate AI Daily Study Guide	
Description	Allows students to receive AI-generated daily study recommendations based on their learning needs and available study time.	
Primary Actor	Student	
Include use cases	None	
Main Flow	Step	Action
	Pre-Condition	Student is logged in and has access to the AI study guide feature.
	1.1	Student navigates to the "AI Daily Study Guide" page.

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:108/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

	1.2	Student enters or selects study-related information such as subjects, tasks, or available time.
	1.3	Student clicks "Generate Guide".
	1.4	System analyzes the input using the AI study planning feature.
	1.5	System generates a personalized daily study guide.
	Post-Condition	Student receives an AI-generated daily study guide successfully.
Alternate Flow	Pre-Condition	Student wants to regenerate the guide with updated information.
	2.1	Student modifies the study input details.
	2.2	Student clicks "Generate Guide" again.
	2.3	System generates an updated daily study guide.
	Post-Condition	Student receives a revised study guide.
Robust Flow	Pre-Condition	Student submits incomplete or empty study input.
	3.1	Student clicks "Generate Guide" without sufficient details.
	3.2	System validates the input.
	3.3	System displays error message: "Please provide enough study information."
	Post-Condition	System prevents invalid guide generation.

Table 55: Use Case Statement – Generate AI Daily Study Guide

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:109/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

4.1.46 Use Case: Analyze Timetable

Use Case ID	UC_046	
Use Case Name	Analyze Timetable	
Description	Allows students to use AI to analyze their timetable and receive suggestions for study planning and schedule improvement.	
Primary Actor	Student	
Include use cases	None	
Main Flow	Step	Action
	Pre-Condition	Student is logged in and has access to the timetable analysis feature.
	1.1	Student navigates to the "AI Timetable Analysis" page.
	1.2	Student enters timetable text or uploads timetable content.
	1.3	Student clicks "Analyze".
	1.4	System processes the timetable using the AI study planning feature.
	1.5	System displays analysis results and suggested study slots.
	Post-Condition	Student receives AI-based timetable analysis successfully.
Alternate Flow	Pre-Condition	Pre-Condition Student wants to analyze another timetable.
	2.1	Student removes the current input.
	2.2	Student enters new timetable text or uploads another timetable.
	2.3	System performs a new analysis.

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:110/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

	Post-Condition	Student receives updated timetable results.
Robust Flow	Pre-Condition	Uploaded timetable is invalid or unreadable.
	3.1	Student uploads incomplete or unsupported timetable content.
	3.2	System fails to process the input correctly.
	3.3	System displays message: "Unable to analyze timetable. Please check the input."
	Post-Condition	System remains stable and informs the user properly.

Table 56: Use Case Statement – Analyze Timetable

4.1.46 Use Case: Generate AI Exam Planner

Use Case ID	UC_046	
Use Case Name	Generate AI Exam Planner	
Description	Allows students to generate an AI-based exam study plan according to exam dates and study needs.	
Primary Actor	Student	
Include use cases	None	
Main Flow	Step	Action
	Pre-Condition	Student is logged in and has exam-related information ready.
	1.1	Student navigates to the "AI Exam Planner" page.
	1.2	Student enters exam subjects, exam dates, and study preferences.
	1.3	Student clicks "Generate Plan".

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:111/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

	1.4	System analyzes the information using the AI exam planning feature.
	1.5	System generates a structured exam study plan.
	Post-Condition	Student receives an AI-generated exam planner successfully.
Alternate Flow	Pre-Condition	Pre-Condition Student wants to adjust the study plan.
	2.1	Student edits the exam information or study preferences.
	2.2	Student clicks "Generate Plan" again.
	2.3	System creates an updated exam plan.
	Post-Condition	Student receives a revised exam study plan.
Robust Flow	Pre-Condition	Pre-Condition Student enters invalid exam data.
	3.1	Student enters incomplete subject names or invalid exam dates.
	3.2	System validates the input.
	3.3	System displays message: "Please enter valid exam information."
	Post-Condition	System prevents invalid exam planner generation.

Table 57: Use Case Statement – Generate AI Exam Planner

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:112/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

4.1.46 Use Case: AI Image Builder

Use Case ID	UC_046	
Use Case Name	AI Image Builder	
Description	Allows users to generate AI educational images or diagrams based on text input.	
Primary Actor	Student, Lecturer	
Include use cases	None	
Main Flow	Step	Action
	Pre-Condition	User is logged in and has access to the AI image generation feature.
	1.1	User navigates to the "AI Image Builder" page.
	1.2	User enters a prompt describing the image or diagram needed.
	1.3	User clicks "Generate Image".
	1.4	System processes the request using the AI image generation feature.
	1.5	System displays the generated educational image.
	Post-Condition	User successfully receives an AI-generated image.
Alternate Flow	Pre-Condition	User wants to generate a different variation.
	2.1	User edits the prompt or request.
	2.2	User clicks "Generate Image" again.
	2.3	System generates a new image variation.
	Post-Condition	User receives an updated AI-generated image.
Robust Flow	Pre-Condition	User submits an empty or unclear prompt.

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:113/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

	3.1	User clicks "Generate Image" without enough description.
	3.2	System validates the prompt.
	3.3	System displays message: "Please enter a clear image description."
	Post-Condition	System prevents invalid image generation.

Table 58: Use Case Statement – AI Image Builder

4.1.46 Use Case: Run AI Plagiarism Detection

Use Case ID	UC_046	
Use Case Name	Run AI Plagiarism Detection	
Description	Allows lecturers to analyze student submissions for possible plagiarism using AI-supported detection.	
Primary Actor	Lecturer	
Include use cases	None	
Main Flow	Step	Action
	Pre-Condition	Lecturer is logged in and student submissions are available.
	1.1	Lecturer opens the assignment submission page.
	1.2	Lecturer selects a student submission.
	1.3	Lecturer clicks "Run AI Plagiarism Detection".
	1.4	System analyzes the submission for possible similarity or flagged content.
	1.5	System displays the plagiarism report and results.

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:114/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

	Post-Condition	Lecturer receives AI-supported plagiarism analysis successfully.
Alternate Flow	Pre-Condition	Lecturer wants to review the report again later.
	2.1	Lecturer saves or reopens the plagiarism report.
	2.2	System loads the stored report.
	Post-Condition	Lecturer reviews the existing plagiarism report.
Robust Flow	Pre-Condition	Submission data is missing or cannot be analyzed
	3.1	Lecturer starts plagiarism detection.
	3.2	System fails to process the selected submission.
	3.3	System displays message: "Unable to analyze submission at this time."
	Post-Condition	System remains stable and no invalid report is produced.

Table 59: Use Case Statement – Run AI Plagiarism Detection

4.1.46 Use Case: AI Grading Assistant

Use Case ID	UC_046
Use Case Name	AI Grading Assistant
Description	Allows lecturers to receive AI-supported grading suggestions and feedback recommendations for student submissions.
Primary Actor	Lecturer

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:115/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

Include use cases	None	
Main Flow	Step	Action
	Pre-Condition	Lecturer is logged in and a student submission is available for review.
	1.1	Lecturer opens a student submission.
	1.2	Lecturer clicks "Use AI Grading Assistant".
	1.3	System analyzes the submission content.
	1.4	System generates suggested marks, comments, or feedback.
	1.5	Lecturer reviews the AI recommendation and applies or edits it.
	Post-Condition	Lecturer receives AI-supported grading assistance successfully.
Alternate Flow	Pre-Condition	Lecturer wants to edit the AI recommendation before saving.
	2.1	Lecturer reviews the suggested grade and feedback.
	2.2	Lecturer modifies the recommended result.
	2.3	Lecturer saves the final grading decision.
	Post-Condition	Submission is graded with lecturer-approved final feedback.
Robust Flow	Pre-Condition	System cannot generate grading recommendation.
	3.1	Lecturer requests AI grading support.
	3.2	System fails to return a recommendation.

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:116/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

	3.3	System displays message: "AI grading recommendation unavailable. Please try again later."
	Post-Condition	System remains stable and lecturer can continue manual grading.

Table 60: Use Case Statement – Use AI Grading Assistant

4.2 Test Case Specification

4.2.1 Test Case: Map Creation Module

Tested By:	Akmal Hakimi bin Abd Rashid
Test Type	Black Box Testing – Functional Testing
Test Design Technique	Use Case Testing
Risk Number	4 (High - Core functionality of the system)
Test Case Number	TC_001.1
Test Case Name	Student: Successfully Create and Save a Mind Map
Requirement Traceability	Student accesses Create Map feature (UC_001)
Test Case Description	To verify that a Student can select a template, edit content, and successfully save a new Thinking Map.
Item(s) to be tested	
1	Map Template Selection Interface
2	Map Editor Canvas (Text Input)
3	Save Functionality
Pre-condition:	
Student is logged into the system and is on the Student Dashboard.	
Specifications	
Input	Expected Output/Result
Student selects "Circle Map" and enters valid title/content.	The system saves the map to the database and redirects the user to the "My Maps" library with the new map listed.
Test Procedure Steps	
1	Log in as a Student using valid credentials.
2	Navigate to the Dashboard and click the Create Map button.
3	Select "Circle Map" from the template list.

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:117/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

4	Click on the center node and enter text: "My Topic".
5	Click on an outer node and enter text: "Detail 1".
6	Click the Save button in the editor toolbar.
7	Verify the system displays a success message "Map Saved".
8	Navigate to "My Maps" library.
9	Verify the new map "My Topic" appears in the list.
Post-condition	
The new Thinking Map is stored in the database and is accessible in the student's library.	

Table 61: Test case map creation module - MySmartStudyApp

Tested By:	Mohamed Azeem bin Sulaiman
Test Type	Black Box Testing – Functional Testing
Test Design Technique	Use Case Testing
Risk Number	2 (Low - Error handling)
Test Case Number	TC_001.2
Test Case Name	Student: Attempt to Save Empty Map
Requirement Traceability	Student accesses Create Map feature (UC_001)
Test Case Description	To verify that the system prevents saving a map without a title or content.
Item(s) to be tested	
1	Save Button Validation Logic
2	Error Message Display
Pre-condition:	
Student is logged in and has opened the Map Editor.	
Specifications	
Input	Expected Output/Result
Student attempts to click "Save" on a blank canvas.	The system blocks the save action and displays an error message requiring input.
Test Procedure Steps	
1	Log in as a Student and open the Create Map interface.
2	Select any template Bubble Map.
3	Leave all text nodes empty.
4	Click the Save button.
5	Verify that the system does not redirect to the library.
6	Verify that an error message "Map content cannot be empty" is displayed.
Post-condition	

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:118/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

No data is written to the database, and the user remains on the Editor screen to correct the input.

Table 62: Test case map creation module - MySmartStudyApp

4.2.2 Test Case: Collaboration Module

Tested By:	Akmal Hakimi bin Abd Rashid
Test Type	Black Box Testing – Functional Testing
Test Design Technique	Use Case Testing
Risk Number	5 (Highest - Real-time complexity and data integrity)
Test Case Number	TC_003.1
Test Case Name	Student: Invite Peer to Collaborate
Requirement Traceability	Student accesses Collaborate on Map feature (UC_003)
Test Case Description	To verify that a student can successfully invite another registered user to view/edit a map.
Item(s) to be tested	
1	Share/Invite Button
2	User Search/Validation
3	Notification System
Pre-condition:	
Two valid student accounts exist (Student A and Student B). Student A has an open map.	
Specifications	
Input	Expected Output/Result
Student A enters Student B's valid email address.	Student B receives an invitation notification and gains access to the map.
Test Procedure Steps	
1	Log in as Student A and open an existing map.
2	Click the Share icon on the toolbar.
3	Enter Student B's registered email address in the input field.
4	Select permission level "Can Edit".
5	Click Send Invite.
6	Log out of Student A and log in as Student B.
7	Check the Notifications or "Shared with Me" section.
8	Verify the invitation from Student A is visible.
Post-condition	
Student B is granted permission to access the specific map ID in the database.	

Table 63: Test case collaboration module - MySmartStudyApp

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:119/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

4.2.3 Test Case: Course Management Module

Tested By:	Mohamed Azeem bin Sulaiman
Test Type	Black Box Testing – Functional Testing
Test Design Technique	Use Case Testing
Risk Number	3 (Medium - Affects student enrollment)
Test Case Number	TC_004.1
Test Case Name	Lecturer: Create New Course and Generate Code
Requirement Traceability	Lecturer accesses Manage Course feature (UC_004)
Test Case Description	To verify that a Lecturer can create a course and the system generates a unique enrollment code.
Item(s) to be tested	
1	Class Creation Form
2	Unique Code Generator Logic
Pre-condition:	
Lecturer is logged into the Web Dashboard.	
Specifications	
Input	Expected Output/Result
Lecturer enters valid Course Name and Subject.	A new Course is created, and a unique 6-digit alphanumeric code is displayed.
Test Procedure Steps	
1	Log in as a Lecturer.
2	Navigate to the Class Management tab.
3	Click Create New Course.
4	Enter Class Name: "BSE Sem 5" and Subject: "Software Testing".
5	Click Create.
6	Verify the new Course appears in the dashboard list.
7	Verify a Course Code, A1B2C3 is generated and visible next to the class name.
Post-condition	
The class is active in the database, and the enrollment code is ready for distribution.	

Table 64: Test Case Course management module - MySmartStudyApp

4.2.4 Test Case: Assignment & Grading Module

Tested By:	Akmal Hakimi bin Abd Rashid
Test Type	Black Box Testing – Functional Testing
Test Design Technique	Use Case Testing
Risk Number	4 (High - Critical for academic workflow)
Test Case Number	TC_005.1
Test Case Name	Lecturer: Publish New Assignment Linked to Map Type

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:120/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

Requirement Traceability	Lecturer accesses Create Assignment feature (UC_005)
Test Case Description	To verify that a Lecturer can successfully create and publish an assignment that requires a specific Thinking Map type.
Item(s) to be tested	
1	Assignment Detail Inputs (Title, Instructions, Date)
2	Map Type Selection Dropdown
3	Publish to Class Logic
Pre-condition:	
Lecturer is logged in and has created at least one active class.	
Specifications	
Input	Expected Output/Result
Lecturer enters assignment details and selects "Double Bubble Map".	The assignment is published, and students in the class receive a notification.
Test Procedure Steps	
1	Log in as a Lecturer and select a specific Class.
2	Click the "Add Assignment" button.
3	Enter Title: "Compare Plant & Animal Cells".
4	Select "Double Bubble Map" from the Map Type dropdown.
5	Set Due Date to a future date.
6	Click "Publish".
7	Verify the assignment appears in the "Active Assignments" list.
8	(Optional) Log in as a Student in that class to verify visibility.
Post-condition	
A new record is added to the Assignments database table linked to the Class ID.	

Table 65: Test Case assignment and grading module - MySmartStudyApp

Tested By:	Mohamed Azeem bin Sulaiman
Test Type	Black Box Testing – Functional Testing
Test Design Technique	Use Case Testing
Risk Number	4 (High - Assessment accuracy)
Test Case Number	TC_006.1
Test Case Name	Lecturer: Grade Submission and Provide Feedback
Requirement Traceability	Lecturer accesses Review & Grade Maps feature (UC_006)

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:121/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

Test Case Description	To verify that a Lecturer can view a student's map, add comments, and assign a grade.
Item(s) to be tested	
1	Submission Viewer
2	Annotation/Comment Tools
3	Grading Input Field
Pre-condition:	
At least one student has submitted a map for an assignment.	
Specifications	
Input	Expected Output/Result
Lecturer enters "85/100" and comment "Good connections".	The grade is saved, and the status updates to "Graded".
Test Procedure Steps	
1	Navigate to the Assignments tab and select an assignment.
2	Click on a student's submission status "Turned In".
3	Verify the student's map loads in the viewer.
4	Use the Comment Tool to click on a node and type "Expand this point".
5	Enter "85" in the Grade field.
6	Click "Return to Student".
7	Verify the status changes from "Turned In" to "Graded".
Post-condition	
Student receives a notification regarding the grade and feedback.	

Table 66: Test Case assignment and grading module - MySmartStudyApp

4.2.5 Test Case: Analytics & Reporting Module

Tested By:	Akmal Hakimi bin Abd Rashid
Test Type	Black Box Testing – Functional Testing
Test Design Technique	Use Case Testing
Risk Number	3 (Medium - Informational)
Test Case Number	TC_007.1
Test Case Name	Lecturer: View Class Performance Analytics
Requirement Traceability	Lecturer accesses View Analytics feature (UC_007)

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:122/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

Test Case Description	To verify that the system correctly calculates and displays visual data regarding student map usage.
Item(s) to be tested	
1	Data Retrieval from Database
2	Chart Rendering (Bar Chart for Map Types)
Pre-condition:	
Class contains students who have created multiple maps.	
Specifications	
Input	Expected Output/Result
Lecturer selects "Analytics" tab for a specific class.	The dashboard loads charts showing "Active Students" and "Map Type Distribution".
Test Procedure Steps	
1	Log in as Lecturer .
2	Navigate to Analytics Dashboard.
3	Select the specific Class from the filter.
4	Verify that the "Most Used Maps" chart is visible (displaying "Circle Map: 40%").
5	Verify that the "Weekly Activity" graph displays data points.
Post-condition	
Analytical data is accurately retrieved and visualized for the user.	

Table 67: Test Case Analytic and Report - MySmartStudyApp

4.2.6 Test Case: Utility Features

Tested By:	Mohamed Azeem bin Sulaiman
Test Type	Black Box Testing – Functional Testing
Test Design Technique	Use Case Testing
Risk Number	3 (Medium - Important for offline submission)
Test Case Number	TC_014.1
Test Case Name	Student: Export Thinking Map to PDF
Requirement Traceability	Student accesses Export Map feature (UC_014)
Test Case Description	To verify that a student can successfully download their digital map as a PDF file.
Item(s) to be tested	
1	Export Button Functionality
2	File Generation & Download
Pre-condition:	
Student has an open map with content.	
Specifications	
Input	Expected Output/Result

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:123/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

Student clicks "Export" and selects "PDF".	A PDF file containing the map image is downloaded to the device.
Test Procedure Steps	
1	Log in as Student and open a saved map.
2	Click the Export icon in the toolbar.
3	Select "PDF Document" from the format options.
4	Confirm the download prompt.
5	Verify that a file named [MapTitle].pdf is saved to the Downloads folder.
6	Open the PDF to verify the map image is clear and legible.
Post-condition	
The file is successfully generated without corrupting the original map data.	

Table 68: Test Case Utility Features - MySmartStudyApp

4.2.7 Test Case: Authentication Module

Tested By:	Akmal Hakimi bin Abd Rashid
Test Type	Black Box Testing – Robustness Testing
Test Design Technique	Boundary Value Analysis / Negative Testing
Risk Number	5 (Highest - Security)
Test Case Number	TC_015.1
Test Case Name	System: Login with Invalid Credentials
Requirement Traceability	User accesses Login feature (UC_015)
Test Case Description	To verify that the system denies access and provides appropriate error messages for incorrect login attempts.
Item(s) to be tested	
1	Authentication Verification Logic
2	Error Message Display
Pre-condition:	
User is on the Login Screen.	
Specifications	
Input	Expected Output/Result
User enters a valid email but an incorrect password.	System denies access and displays "Invalid email or password".
Test Procedure Steps	
1	Navigate to the Login page.
2	Enter a registered email (e.g., student@gmail.com).
3	Enter an incorrect password (e.g., WrongPass123).
4	Click Login.
5	Verify that the dashboard does not load.
6	Verify that the error message "Invalid email or password" is visible.

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:124/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

Post-condition

User remains on the login screen, and the failed attempt is logged (internally).

Table 69: Test Case Authentication Model – MySmartStudyApp

4.2.8 Test Case: User Profile Module

Tested By:	Akmal Hakimi bin Abd Rashid
Test Type	Black Box Testing – Functional Testing
Test Design Technique	State Transition Testing
Risk Number	3 (Medium - Data accuracy and personalization)
Test Case Number	TC_016.1
Test Case Name	Student: Edit Profile Information
Requirement Traceability	Student accesses Profile Settings feature (UC_016)
Test Case Description	To verify that a student can successfully update their personal details (e.g., Phone Number, Display Name).
Item(s) to be tested	
1	"Edit Profile" Button
2	Input Fields (Name, Phone, Bio)
3	Save/Cancel Controls
Pre-condition:	
Student A is logged in and is currently on the Dashboard page.	
Specifications	
Input	Expected Output/Result
Student updates the "Phone Number" field with a valid format and clicks Save.	System saves the new data, updates the database, and displays a "Profile Updated" success message.
Test Procedure Steps	
1	Log in as Student A and navigate to the User Profile section.
2	Click the Edit Profile button.
3	Clear the existing phone number in the "Phone" field.
4	Enter a new valid phone number (e.g., +60123456789).
5	Click the Save Changes button.
6	Refresh the page or navigate away and come back.
7	Verify that the "Phone" field still displays the new number.
Post-condition	
Student A's profile record is updated in the database.	

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:125/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

Table 70: Test Case User Profile Model – MySmartStudyApp

4.2.9 Test Case: Dashboard Management Module

Tested By:	Akmal Hakimi bin Abd Rashid
Test Type	Black Box Testing – Functional Testing
Test Design Technique	Use Case Testing
Risk Number	4 (High - Data loss prevention)
Test Case Number	TC_017.1
Test Case Name	Student: Delete an Existing Map
Requirement Traceability	Student accesses Delete Map option (UC_017)
Test Case Description	To verify that a student can permanently delete a mind map from their library.
Item(s) to be tested	
1	Map Context Menu (Three dots icon)
2	Delete Option
3	Confirmation Modal (Yes/No)
Pre-condition:	
Student A has at least one existing map titled "Test Map 1" on the dashboard.	
Specifications	
Input	Expected Output/Result
Student selects "Delete" and confirms the action in the popup.	The map is removed from the dashboard list and the database record is deleted.
Test Procedure Steps	
1	Log in as Student A and locate "Test Map 1" on the dashboard.
2	Click the Options (three dots) icon on the map card.
3	Select Delete from the dropdown menu.
4	Observe the confirmation popup warning about permanent data loss.
5	Click Confirm / Yes.
6	Verify that "Test Map 1" is no longer visible on the dashboard.
7	Check the "Trash" or "Archived" section (if applicable) to ensure it is gone.
Post-condition	
Map ID is removed from the active database; associated nodes are deleted.	

Table 71: Test Case Dashboard Management Model – MySmartStudyApp

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:126/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

4.2.10 Test Case: Search & Navigation Module

Tested By:	Akmal Hakimi bin Abd Rashid
Test Type	Black Box Testing – Functional Testing
Test Design Technique	Boundary Value Analysis
Risk Number	2 (Low - User convenience)
Test Case Number	TC_018.1
Test Case Name	Student: Filter Maps by Category
Requirement Traceability	Student accesses Filter/Sort feature (UC_018)
Test Case Description	To verify that the dashboard can filter displayed maps based on assigned categories (e.g., Subject, Date).
Item(s) to be tested	
1	Filter Dropdown Menu
2	Map List View
3	Category Tags
Pre-condition:	
Student A has maps tagged with different categories Science or History.	
Specifications	
Input	Expected Output/Result
Student selects "Science" from the filter dropdown.	The dashboard refreshes to show ONLY maps tagged with "Science".
Test Procedure Steps	
1	Log in as Student A and view the main dashboard.
2	Locate the Filter By dropdown menu near the search bar.
3	Click the dropdown and select "Science".
4	Verify that the list updates immediately.
5	Check that all visible map cards have the "Science" tag.
6	Change the filter back to "All".
7	Verify that all maps (Science and History) are visible again.
Post-condition	
View state is updated; no data modification occurs in the database.	

Table 72: Test Case Search and Navigation – MySmartStudyApp

4.2.11 Test Case: Map Visual Customization Module

Tested By:	Akmal Hakimi Bin Abd Rashid
Test Type	Black Box Testing – Functional Testing

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:127/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

Test Design Technique	Use Case Testing
Risk Number	3 (Medium - User customization and UI stability)
Test Case Number	TC_019.1
Test Case Name	Student: Customize Node Color and Style
Requirement Traceability	Student accesses Node Formatting feature (UC_019)
Test Case Description	To verify that a student can change the background color and shape of a specific node within the mind map.
Item(s) to be tested	
1	Node Selection Interface
2	Color Palette Picker
3	Shape Dropdown Menu
Pre-condition:	
Student A has a map open with at least one active node selected.	
Specifications	
Input	Expected Output/Result
Student selects a node and chooses "Red" from the color palette.	The specific node's background changes to Red immediately without affecting other nodes.
Test Procedure Steps	
1	Log in as Student A and open an existing map.
2	Click on a "Child Node" to select it (highlight border appears).
3	Click the Format/Style icon in the toolbar.
4	Open the Color Palette and select the color Red (#FF0000).
5	Verify the node color updates instantly.
6	From the same menu, change the Shape from "Rectangle" to "Oval".
7	Verify the node boundary changes to an oval shape.
8	Click Save to ensure changes persist.
Post-condition	
The visual style parameters for the specific node are updated in the JSON/XML map data.	

Table 73: Test Case Map Visual Customization Model – MySmartStudyApp

4.2.12 Test Case: Help and Support Module

Tested By:	Mohamed Azeem bin Sulaiman
Test Type	Black Box Testing – Functional Testing
Test Design Technique	Boundary Value Analysis
Risk Number	2 (Low - Support functionality)
Test Case Number	TC_020.1
Test Case Name	Student: Submit Feedback/Support Ticket

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:128/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

Requirement Traceability	Student accesses Help Center (UC_020)
Test Case Description	To verify that the user can successfully send a text-based inquiry or bug report to the admin team.
Item(s) to be tested	
1	"Contact Us" Form
2	Message Text Area (Character limit validation)
3	Submit Button
Pre-condition:	
Student is logged in. The internet connection is active.	
Specifications	
Input	Expected Output/Result
Student fills in the subject and description, then clicks Submit.	System sends the data to the support backend and displays a "Message Sent" confirmation.
Test Procedure Steps	
1	Log in as Student A and navigate to Settings > Help.
2	Select "Report a Bug" from the category list.
3	Enter "Map failed to load" in the Subject line.
4	Enter a description (min 10 characters, max 500 characters).
5	Click the Submit Ticket button.
6	Observe the screen for a success toast notification/popup.
7	Check the student's registered email for an automated receipt acknowledgment.
Post-condition	
A new support ticket record is created in the Admin Support Database.	

Table 74: Test Case Help and Support Model – MySmartStudyApp

4.2.13 Test Case: Session Management Module

Tested By:	Mohamed Azeem bin Sulaiman
Test Type	Black Box Testing – Security Testing
Test Design Technique	State Transition Testing
Risk Number	4 (High - Security and Account Access)
Test Case Number	TC_021.1
Test Case Name	Student: System Logout
Requirement Traceability	Student accesses Logout function (UC_021)
Test Case Description	To verify that the "Logout" function securely clears the session and prevents unauthorized back-navigation.

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:129/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

Item(s) to be tested	
1	Logout Button
2	Browser Session/Cookies
3	Browser "Back" Button
Pre-condition:	
Student A is currently logged in and viewing the Dashboard.	
Specifications	
Input	Expected Output/Result
Student clicks "Logout".	System destroys the session token and redirects to the Login Page.
Test Procedure Steps	
1	Log in as Student A.
2	Click on the Profile Avatar in the top right corner.
3	Select Logout from the dropdown menu.
4	Verify the user is redirected to the Login/Landing Page.
5	Attempt to click the browser's Back button to return to the dashboard.
6	Verify that the system does not load the previous page but redirects to Login.
Post-condition	
Session token is invalidated on the server. User must re-authenticate to access the system.	

Table 75: Test Case Session management Model – MySmartStudyApp

4.2.14 Test Case: Multimedia Attachment Module

Tested By:	Akmal Hakimi Bin Abd Rashid
Test Type	Black Box Testing – Functional Testing
Test Design Technique	Use Case Testing
Risk Number	3 (Medium - Storage and display performance)
Test Case Number	TC_022.1
Test Case Name	Student: Insert Image into Node
Requirement Traceability	Student accesses Image Upload feature (UC_022)
Test Case Description	To verify that a student can upload and attach an image file (JPG/PNG) to a specific node on the mind map.
Item(s) to be tested	
1	Node Context Menu
2	File Browser/Uploader
3	Image Rendering Container

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:130/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

Pre-condition:	
Student A has an open map and a valid image file (under 5MB) stored on their device.	
Specifications	
Input	Expected Output/Result
Student selects "Attach Image" and uploads a valid PNG file.	The image is uploaded, processed, and displayed inside or attached to the selected node.
Test Procedure Steps	
1	Log in as Student A and open a map.
2	Right-click on a target node to open the context menu.
3	Select "Insert Image" or "Attachment".
4	Browse the local computer files and select a .png image file.
5	Click Open/Upload.
6	Wait for the upload progress to complete.
7	Verify the image appears attached to the node.
8	Resize the node to ensure the image scales correctly.
Post-condition	
The image file is stored in the cloud storage bucket and linked to the node ID in the database.	

Table 76: Test Case Multimedia attachment Model – MySmartStudyApp

4.2.15 Test Case: Real-Time Collaboration Module

Tested By:	Mohamed Azeem bin Sulaiman
Test Type	Black Box Testing – Integration/Sync Testing
Test Design Technique	State Transition Testing
Risk Number	5 (Highest - Data integrity during concurrent access)
Test Case Number	TC_023.1
Test Case Name	Student: View Peer Edits in Real-Time
Requirement Traceability	Student accesses Shared Map Session (UC_023)
Test Case Description	To verify that changes made by one collaborator are immediately visible to another collaborator without refreshing the page.
Item(s) to be tested	
1	WebSocket Connection
2	Canvas Refresh Rate
3	User Presence Indicator
Pre-condition:	
Student A and Student B are both logged in and viewing the <i>same</i> shared map simultaneously on different devices/browsers.	
Specifications	

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:131/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

Input	Expected Output/Result
Student B adds a new node titled "Collaboration Test".	Student A sees the new node appear on their screen instantly (within 2 seconds).
Test Procedure Steps	
1	Log in as Student A (Browser 1) and open the shared map.
2	Log in as Student B (Browser 2) and open the same map.
3	On Student B's screen, create a new child node and type "Hello World".
4	Immediately look at Student A's screen.
5	Verify that the node "Hello World" appears automatically without clicking refresh.
6	Verify that a "typing" or "presence" indicator shows Student B is editing.
Post-condition	
Map state is synchronized across all active sessions.	

Table 77: Test Case Real Time Collaboration Model – MySmartStudyApp

4.2.16 Test Case: History & Restoration Module

Tested By:	Akmal Hakimi Bin Abd Rashid
Test Type	Black Box Testing – Functional Testing
Test Design Technique	State Transition Testing
Risk Number	3 (Medium - User Experience and Data Recovery)
Test Case Number	TC_024.1
Test Case Name	Student: Undo Last Action
Requirement Traceability	Student accesses Undo/Redo feature (UC_024)
Test Case Description	To verify that a student can reverse their most recent editing action (e.g., accidental deletion or text change) using the Undo function.
Item(s) to be tested	
1	Undo Button (Toolbar)
2	Keyboard Shortcut (Ctrl+Z)
3	Map State Buffer
Pre-condition:	
Student A is editing a map and has performed at least one action (e.g., moved a node).	
Specifications	
Input	Expected Output/Result
Student moves Node A to a new position,	Node A reverts to its original position instantly.

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:132/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

then immediately clicks "Undo".	
Test Procedure Steps	
1	Log in as Student A and open an existing map.
2	Select a specific node (e.g., "Main Topic").
3	Drag and drop the node to a different location on the canvas.
4	Locate the Undo Arrow icon in the top toolbar.
5	Click the Undo icon (or press Ctrl+Z).
6	Verify that the node jumps back to its exact previous coordinates.
7	Click the Redo icon.
8	Verify the node moves back to the new location.
Post-condition	
Map state is restored to the previous timestamp in the session history.	

Table 78: Test Case History and Restoration Model – MySmartStudyApp

4.2.17 Test Case: Tagging & Organization Module

Tested By:	Mohamed Azeem bin Sulaiman
Test Type	Black Box Testing – Functional Testing
Test Design Technique	Use Case Testing
Risk Number	2 (Low - Organization convenience)
Test Case Number	TC_025.1
Test Case Name	Student: Add Tags to Map
Requirement Traceability	Student accesses Map Settings (UC_025)
Test Case Description	To verify that a student can assign custom tags (e.g., "Exam", "Biology") to a map for easier filtering.
Item(s) to be tested	
1	Map Details Panel
2	"Add Tag" Input Field
3	Save Button
Pre-condition:	
Student A has created a map and is viewing it on the Dashboard.	
Specifications	
Input	Expected Output/Result
Student types "Final Exam" into the tag field and saves.	The tag "Final Exam" appears on the map card and can be used for searching.
Test Procedure Steps	

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:133/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

1	Log in as Student A.
2	Locate a map card on the Dashboard.
3	Click the Settings/Info icon on the map card.
4	Locate the "Tags" section.
5	Type "Final Exam" and press Enter.
6	Verify the tag appears as a "chip" or label below the title.
7	Refresh the page.
8	Verify the tag persists on the map card.
Post-condition	
The tag string is associated with the Map ID in the database.	

Table 79: Test Case Tagging and organization Model – MySmartStudyApp

4.2.18 Test Case: Auto-Save Module

Tested By:	Akmal Hakimi Bin Abd Rashid
Test Type	Black Box Testing – Functional/Reliability Testing
Test Design Technique	Experience-based Testing
Risk Number	5 (Highest - Data Integrity)
Test Case Number	TC_026.1
Test Case Name	Student: Verify Auto-Save Functionality
Requirement Traceability	System performs Background Save (UC_026)
Test Case Description	To verify that the system automatically saves changes to the cloud without the user manually clicking a "Save" button.
Item(s) to be tested	
1	Save Status Indicator (e.g., "Saving...", "Saved")
2	Database Connectivity
Pre-condition:	
Student A is editing a map. Internet connection is stable.	
Specifications	
Input	Expected Output/Result
Student adds a node and closes the browser tab immediately after the "Saved" indicator appears.	Upon re-opening the map, the new node is present.
Test Procedure Steps	
1	Log in as Student A and open a map.
2	Add a new node with the text " Auto-Save Test ".
3	Observe the top status bar changing from "Saving..." to " All changes saved ".

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:134/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

4	Immediately close the browser tab or window.
5	Open a new browser window and log in again.
6	Open the same map.
7	Verify that the node " Auto-Save Test " is visible.
Post-condition	
Data remains consistent between the client and server.	

Table 80: Test Case Auto Save Model – MySmartStudyApp

4.2.19 Test Case: Canvas Navigation Module

Tested By:	Mohamed Azeem bin Sulaiman
Test Type	Black Box Testing – Functional Testing
Test Design Technique	Boundary Value Analysis
Risk Number	3 (Medium - Usability)
Test Case Number	TC_027.1
Test Case Name	Student: Zoom and Pan Canvas
Requirement Traceability	Student interacts with Canvas Tools (UC_027)
Test Case Description	To verify that the user can zoom in/out and pan across large maps to view different sections.
Item(s) to be tested	
1	Zoom Control (+ / - buttons)
2	Mouse Wheel Scroll
3	Canvas Drag Interaction
Pre-condition:	
Student A has a large map open that extends beyond the visible screen area.	
Specifications	
Input	Expected Output/Result
Student clicks "Zoom Out" (-) until 50%.	The map scales down, showing more content in a smaller size.
Test Procedure Steps	
1	Log in as Student A and open a large map.
2	Click the Zoom Out (-) button in the bottom corner.
3	Verify the zoom percentage indicator drops (e.g., 100% -> 90% -> 80%).
4	Click and hold the left mouse button on an empty area of the canvas.
5	Drag the mouse (Pan action).
6	Verify the view shifts to reveal hidden parts of the map.
7	Click the "Fit to Screen" button (if available).
8	Verify the entire map centers within the viewport.

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:135/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

Post-condition
Viewport coordinates are updated temporarily; actual map data is unchanged.

Table 81: Test Case Canvas Navigation Model – MySmartStudyApp

4.2.20 Test Case: Presentation Module

Tested By:	Akmal Hakimi Bin Abd Rashid
Test Type	Black Box Testing – Functional Testing
Test Design Technique	State Transition Testing
Risk Number	3 (Medium - Feature Utility)
Test Case Number	TC_028.1
Test Case Name	Student: Enter Presentation Mode
Requirement Traceability	Student accesses Present Map feature (UC_028)
Test Case Description	To verify that the student can switch the view to a full-screen "Presentation Mode" to display the map like a slideshow.
Item(s) to be tested	
1	Presentation Button
2	Full-screen Toggle
3	Navigation Controls (Next/Previous)
Pre-condition:	
Student A has a completed map open in the editor.	
Specifications	
Input	Expected Output/Result
Student clicks the "Present" button.	The interface hides toolbars, enters full-screen, and centers the root node.
Test Procedure Steps	
1	Log in as Student A and open a map.
2	Click the Present / Slideshow icon in the toolbar.
3	Verify that all editing toolbars (sidebars/top bars) disappear.
4	Verify the map takes up the entire screen space.
5	Press the Right Arrow Key on the keyboard.
6	Verify the camera focuses on the first child node.
7	Press Esc to exit Presentation Mode.
Post-condition	
System returns to the standard editing interface.	

Table 82: Test Case Presentation Model – MySmartStudyApp

4.2.21 Test Case: Mobile Responsiveness Module

Tested By:	Mohamed Azeem bin Sulaiman
-------------------	-----------------------------------

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:136/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

Test Type	Black Box Testing – UI/UX Testing
Test Design Technique	Equivalence Partitioning
Risk Number	4 (High - Accessibility across devices)
Test Case Number	TC_029.1
Test Case Name	Student: Access Map on Mobile View
Requirement Traceability	System adapts to Mobile Layout (UC_029)
Test Case Description	To verify that the application interface adjusts correctly when accessed via a mobile browser or small screen resolution.
Item(s) to be tested	
1	Responsive Layout Grid
2	Hamburger Menu
3	Touch Interaction (Tap to Select)
Pre-condition:	
Student A is logged in. Browser window is resized to mobile width (approx. 375px) or accessed via a smartphone.	
Specifications	
Input	Expected Output/Result
Student resizes the browser window to <600px width.	The sidebar collapses into a hamburger menu, and buttons scale down to fit the screen.
Test Procedure Steps	
1	Log in as Student A on a desktop browser.
2	Open Developer Tools (F12) and toggle Device Toolbar (Mobile View).
3	Select iPhone SE or Pixel 5 dimensions.
4	Verify that the top navigation bar condenses.
5	Verify that the "Tools" panel moves to a bottom sheet or hidden menu.
6	Attempt to create a node using the (+) floating action button.
7	Verify the node is created successfully in the center.
Post-condition	
UI elements are rearranged; functional logic remains the same.	

Table 83: Test Case Mobile Responsiveness Model – MySmartStudyApp

4.2.22 Test Case: Keyboard Shortcut Module

Tested By:	Akmal Hakimi Bin Abd Rashid
Test Type	Black Box Testing – Functional Testing
Test Design Technique	Use Case Testing
Risk Number	2 (Low - Efficiency feature)

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:137/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

Test Case Number	TC_030.1
Test Case Name	Student: Quick Node Creation via Hotkeys
Requirement Traceability	Student uses Hotkeys (UC_030)
Test Case Description	To verify that power users can create nodes using standard keyboard shortcuts (Tab and Enter) instead of mouse clicks.
Item(s) to be tested	
1	Keyboard Event Listener
2	Node Creation Logic
Pre-condition:	
Student A has a map open and a "Parent Node" is currently selected.	
Specifications	
Input	Expected Output/Result
Student presses the "Tab" key.	A new Child Node is created and automatically attached to the selected Parent Node.
Test Procedure Steps	
1	Log in as Student A and open a new blank map.
2	Click on the Central Topic to select it.
3	Press the Tab key on the keyboard.
4	Verify a "Subtopic" node appears.
5	Type "Test Node" and press Enter .
6	Verify the text is saved and a new "Sibling Node" (same level) is created below it.
7	Press Delete or Backspace .
8	Verify the newly created sibling node is deleted.
Post-condition	
Map structure is updated in memory and synced to the database.	

Table 84: Test Case Keyboard Shortcut Model – MySmartStudyApp

4.2.23 Test Case: Commenting Module

Tested By:	Akmal Hakimi Bin Abd Rashid
Test Type	Black Box Testing – Functional Testing
Test Design Technique	Use Case Testing
Risk Number	3 (Medium - Collaboration feature)
Test Case Number	TC_031.1
Test Case Name	Student: Add Comment to Node
Requirement Traceability	Student accesses Node Comments (UC_032)
Test Case Description	To verify that a student can leave a textual comment on a specific node for personal notes or peer feedback.
Item(s) to be tested	

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:138/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

1	Comment Icon/Button
2	Text Input Box
3	Comment Badge Indicator
Pre-condition:	
Student A has a map open. If collaborating, Student B acts as the viewer.	
Specifications	
Input	Expected Output/Result
Student selects a node, types a comment, and clicks Post.	A small comment badge/icon appears on the node indicating a note exists.
Test Procedure Steps	
1	Log in as Student A and open a map.
2	Select a specific node.
3	Click the "Add Comment" icon in the toolbar or context menu.
4	Type "Review this section later" in the input field.
5	Click Post.
6	Verify a small "bubble" or "1" badge appears on the corner of the node.
7	Click the badge to expand and read the comment.
Post-condition	
The comment data is stored and linked to the Node ID and User ID.	

Table 85:Test Case Commenting Model – MySmartStudyApp

4.2.24 Test Case: Template Library Module

Tested By:	Mohamed Azeem bin Sulaiman
Test Type	Black Box Testing – Functional Testing
Test Design Technique	State Transition Testing
Risk Number	3 (Medium - Content creation efficiency)
Test Case Number	TC_032.1
Test Case Name	Student: Apply System Template
Requirement Traceability	Student accesses Template Library (UC_033)
Test Case Description	To verify that a student can start a new map using a pre-defined structure (e.g., Fishbone Diagram) instead of a blank canvas.
Item(s) to be tested	
1	"Create New" Modal
2	Template Selection Grid
3	Canvas Initialization Logic
Pre-condition:	
Student A is on the Dashboard.	

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:139/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

Specifications	
Input	Expected Output/Result
Student selects the "Fishbone" template during creation.	The editor opens with a pre-populated Fishbone structure (Head + Spines) instead of a single central node.
Test Procedure Steps	
1	Log in as Student A.
2	Click the (+) Create New Map button.
3	Scroll through the template list and select "Fishbone Diagram".
4	Click Create.
5	Wait for the editor to load.
6	Verify that the map is not blank and contains the standard Fishbone layout nodes (Causes/Effect).
7	Edit the text of the "Main Effect" node.
Post-condition	
A new map is created with pre-inserted node relationships based on the template schema.	

Table 86: Test Case Template Library Model – MySmartStudyApp

4.2.25 Test Case: Printing Module

Tested By:	Akmal Hakimi Bin Abd Rashid
Test Type	Black Box Testing – Functional Testing
Test Design Technique	Use Case Testing
Risk Number	2 (Low - Utility feature)
Test Case Number	TC_033.1
Test Case Name	Student: Print Map Layout
Requirement Traceability	Student accesses Print function (UC_034)
Test Case Description	To verify that the user can send the current map view to a printer with correct formatting and layout settings.
Item(s) to be tested	
1	Print Preview Modal
2	Page Setup (A4/Letter)
3	Browser Print Dialog Integration
Pre-condition:	
Student A has a map open. A local printer is connected or "Save as PDF" driver is available.	
Specifications	
Input	Expected Output/Result
Student selects "Print" and chooses	The system opens the browser print dialog with the map scaled to fit the selected page size.

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:140/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

Landscape orientation.	
Test Procedure Steps	
1	Log in as Student A and open a map.
2	Click the File menu or Export icon.
3	Select Print.
4	Verify a "Print Preview" window appears showing the map.
5	Adjust settings to Landscape mode.
6	Click Print.
7	Verify the system hands off the data to the native OS/Browser print dialog.
Post-condition	
No changes to the database. A print job is created on the local machine.	

Table 87: Test Case Printing Model – MySmartStudyApp

4.2.26 Test Case: Spell Check Module

Tested By:	Mohamed Azeem bin Sulaiman
Test Type	Black Box Testing – Functional Testing
Test Design Technique	Error Guessing
Risk Number	2 (Low - Content Quality)
Test Case Number	TC_034.1
Test Case Name	Student: Verify Typo Detection
Requirement Traceability	System performs Spell Check (UC_035)
Test Case Description	To verify that the system flags misspelled words within node text and offers corrections.
Item(s) to be tested	
1	Text Input Editor
2	Red Underline Indicator
3	Context Menu Suggestions
Pre-condition:	
Student A has a map open. Browser spell-check or app-integrated dictionary is active.	
Specifications	
Input	Expected Output/Result
Student types "Envirnoment"	The word is underlined in red. Right-clicking offers "Environment" as a correction.

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:141/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

(incorrect) into a node.	
Test Procedure Steps	
1	Log in as Student A and create a new node.
2	Type the misspelled word " Envirnoment " and press Space.
3	Verify that a red wavy line appears under the word.
4	Right-click on the word.
5	Verify that the correct spelling " Environment " is suggested.
6	Click the suggestion.
7	Verify the text is replaced and the red line disappears.
Post-condition	
Node text is updated in memory and pending save.	

Table 88: Test Case Spell Check Model – MySmartStudyApp

4.2.27 Test Case: Trash & Recovery Module

Tested By:	Akmal Hakimi Bin Abd Rashid
Test Type	Black Box Testing – Functional Testing
Test Design Technique	State Transition Testing
Risk Number	3 (Medium - Data Safety)
Test Case Number	TC_035.1
Test Case Name	Student: Restore Deleted Map
Requirement Traceability	Student accesses Trash Bin (UC_036)
Test Case Description	To verify that a student can recover a map that was accidentally deleted within the allowable retention period (e.g., 30 days).
Item(s) to be tested	
1	Trash/Bin Section
2	Restore Button
3	Dashboard Map List
Pre-condition:	
Student A has previously deleted a map (see TC_017.1) and it currently resides in the Trash.	
Specifications	
Input	Expected Output/Result
Student selects the deleted map in Trash and clicks "Restore".	The map is moved from Trash back to the main Dashboard and is editable again.
Test Procedure Steps	
1	Log in as Student A .
2	Navigate to the Trash or Bin tab in the sidebar.
3	Locate the map deleted in the previous test session.

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:142/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

4	Click the Restore / Recover icon on the map card.
5	Verify the map disappears from the Trash list.
6	Navigate back to the My Maps (Main Dashboard).
7	Verify the map is visible and can be opened.
Post-condition	
Map status in the database is updated from "Deleted" to "Active".	

Table 89: Test Case Trash and Recovery Model – MySmartStudyApp

4.2.28 Test Case: Offline Mode Handling Module

Tested By:	Mohamed Azeem bin Sulaiman
Test Type	Black Box Testing – Functional/Recovery Testing
Test Design Technique	Use Case Testing
Risk Number	4 (High - Data loss prevention)
Test Case Number	TC_036.1
Test Case Name	Student: Edit Map Without Internet
Requirement Traceability	System handles Network Disconnection (UC_037)
Test Case Description	To verify that the application allows limited editing when the internet connection is lost and syncs when reconnected.
Item(s) to be tested	
1	Network Status Indicator
2	Local Browser Storage
3	Sync Mechanism
Pre-condition:	
Student A is logged in and editing a map. Internet connection is active initially.	
Specifications	
Input	Expected Output/Result
Student disconnects internet, adds a node, then reconnects.	System shows "Offline" warning, saves locally, and auto-syncs when online.
Test Procedure Steps	
1	Log in as Student A and open a map.
2	Manually turn off the Wi-Fi/Internet connection on the device.
3	Verify a banner appears saying "You are offline. Changes saved locally."
4	Create a new node with text "Offline Node" .
5	Turn the Wi-Fi/Internet connection back on.
6	Verify the banner changes to "Back Online" or "Synced" .
7	Refresh the page to ensure the "Offline Node" is still present.
Post-condition	

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:143/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

Local changes are pushed to the server database upon reconnection.

Table 90: Test Case Offline Mode and Handling Model – MySmartStudyApp

4.2.29 Test Case: Cross-Map Linking Module

Tested By:	Akmal Hakimi Bin Abd Rashid
Test Type	Black Box Testing – Functional Testing
Test Design Technique	State Transition Testing
Risk Number	3 (Medium - Navigation complexity)
Test Case Number	TC_037.1
Test Case Name	Student: Link Node to Another Map
Requirement Traceability	Student accesses Hyperlink Feature (UC_038)
Test Case Description	To verify that a student can create a hyperlink on a node that redirects to a different mind map in their library.
Item(s) to be tested	
1	"Add Link" Menu
2	Map Selection Picker
3	Hyperlink Icon/Behavior
Pre-condition:	
Student A has two maps: Map 1 (Open) and Map 2 (Target).	
Specifications	
Input	Expected Output/Result
Student selects a node in Map 1 and links it to Map 2.	Clicking the node in Map 1 opens Map 2 in a new tab or window.
Test Procedure Steps	
1	Log in as Student A and open Map 1.
2	Right-click a node and select "Add Hyperlink".
3	Select "Internal Map" tab in the popup.
4	Choose Map 2 from the list of available maps.
5	Click Insert.
6	Verify a "Link" icon appears on the node.
7	Click the icon/node.
8	Verify that Map 2 opens successfully.
Post-condition	
A directional link relationship is established between Node ID (Map 1) and Map ID (Map 2).	

Table 91: Test Case Cross Map Lingking Model – MySmartStudyApp

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:144/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

4.2.30 Test Case: Drag and Drop Organization Module

Tested By:	Akmal Hakimi Bin Abd Rashid
Test Type	Black Box Testing – UI/Functional Testing
Test Design Technique	Use Case Testing
Risk Number	3 (Medium - Usability)
Test Case Number	TC_038.1
Test Case Name	Student: Reorganize Nodes via Drag-and-Drop
Requirement Traceability	Student interacts with Node structure (UC_040)
Test Case Description	To verify that a student can change the parent-child relationship of a node by dragging it to a different branch.
Item(s) to be tested	
1	Mouse Drag Event
2	Drop Target Highlight
3	Auto-Layout Adjustment
Pre-condition:	
Student A has a map with at least two different main branches (Branch A and Branch B).	
Specifications	
Input	Expected Output/Result
Student drags a child node from Branch A and drops it onto Branch B.	The node disconnects from Branch A and attaches to Branch B automatically.
Test Procedure Steps	
1	Log in as Student A and open a map.
2	Click and hold the left mouse button on a Child Node in Branch A.
3	Drag the mouse cursor towards Branch B.
4	Verify that Branch B highlights/glows to indicate it is a valid target.
5	Release the mouse button (Drop).
6	Verify the Child Node is now attached to Branch B.
7	Verify the connector line is redrawn correctly.
Post-condition	
The node's parent_id is updated in the database structure.	

Table 92: Test Case Drag and Drop Organization Model – MySmartStudyApp

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:145/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

4.2.31 Test Case: Quiz Module

Tested By:	Akmal Hakimi Bin Abd Rashid
Test Type	Black Box Testing – Functional Testing
Test Design Technique	Use Case Testing
Risk Number	4 (High - Important academic workflow)
Test Case Number	TC_047.1
Test Case Name	Student: Successfully Attempt and Submit Quiz
Requirement Traceability	Student accesses Take Quiz feature (UC_047 or Quiz Use Case)
Test Case Description	To verify that a student can access, answer, and submit a quiz successfully.
Item(s) to be tested	
1	Quiz Access Page
2	Question Display Interface
3	Submit Quiz Functionality
Pre-condition:	
Student is logged into the system and enrolled in a course with an available quiz.	
Specifications	
Input	Expected Output/Result
Student selects a quiz, answers all questions, and clicks Submit.	The system saves the quiz attempt and displays the result or submission confirmation successfully.
Test Procedure Steps	
1	Log in as a Student using valid credentials.
2	Navigate to the selected course.
3	Open the Quiz section.
4	Select an available quiz.
5	Answer all quiz questions.
6	Click the Submit button.
7	Verify the system displays a success message or result page.
8	Verify the quiz attempt is recorded.
Post-condition	
The quiz attempt is stored in the system and is accessible in the student's quiz records.	

Table 81: Test Case Quiz Module – MySmartStudyApp

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:146/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

4.2.32 Test Case: Messaging Module

Tested By:	Mohamed Azeem bin Sulaiman
Test Type	Black Box Testing – Functional Testing
Test Design Technique	Use Case Testing
Risk Number	3 (Medium - Communication feature)
Test Case Number	TC_048.1
Test Case Name	User: Send Private Message Successfully
Requirement Traceability	User accesses Send Private Message feature (Messaging Use Case)
Test Case Description	To verify that a user can send a private message to another user successfully.
Item(s) to be tested	
1	Message Input Box
2	Send Button
3	Message Delivery and Display
Pre-condition:	
Two valid user accounts exist and both users are registered in the system.	
Specifications	
Input	Expected Output/Result
User sends a valid text message to another user.	The message is delivered successfully and appears in the conversation thread.
Test Procedure Steps	
1	Log in as User A.
2	Navigate to the Messages page.
3	Open an existing conversation or start a new one.
4	Type a valid message such as “Hello, please check the assignment.”
5	Click Send.
6	Log in as User B or refresh User B’s conversation page.
7	Verify the message appears correctly in the conversation.
Post-condition	
The private message is stored in the system and visible to both users.	

Table 82: Test Case Messaging Module – MySmartStudyApp

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:147/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

4.2.33 Test Case: Notification Module

Tested By:	Akmal Hakimi Bin Abd Rashid
Test Type	Black Box Testing – Functional Testing
Test Design Technique	Use Case Testing
Risk Number	4 (High - Important academic workflow)
Test Case Number	TC_049.1
Test Case Name	User: View Notifications Successfully
Requirement Traceability	User accesses View Notifications feature (Notification Use Case)
Test Case Description	To verify that the system correctly displays notifications for the logged-in user.
Item(s) to be tested	
1	Notification List View
2	Read/Unread Status
3	Notification Detail Access
Pre-condition:	
User is logged in and at least one notification exists in the account.	
Specifications	
Input	Expected Output/Result
User opens the Notifications page.	The system displays the correct list of notifications for that user.
Test Procedure Steps	
1	Log in as a valid user.
2	Navigate to the Notifications page.
3	Verify that the notification list loads.
4	Click one notification item.
5	Verify that the notification content is displayed correctly.
6	Mark the notification as read if applicable.
7	Verify the notification status updates.
Post-condition	
The selected notification is displayed correctly and its status is updated if read.	

Table 84: Test Case Notification Module – MySmartStudyApp

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:148/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

4.2.34 Test Case: Attendance Module

Tested By:	Akmal Hakimi Bin Abd Rashid
Test Type	Black Box Testing – Functional Testing
Test Design Technique	Use Case Testing
Risk Number	3 (Medium - Academic record feature)
Test Case Number	TC_050.1
Test Case Name	User: View Attendance Successfully
Requirement Traceability	User accesses View Attendance feature (Attendance Use Case)
Test Case Description	To verify that attendance records are displayed correctly for the selected course or user.
Item(s) to be tested	
1	Attendance Page
2	Attendance Record Display
3	Course/Session Selection
Pre-condition:	
User is logged in and attendance records exist in the system.	
Specifications	
Input	Expected Output/Result
User opens the Attendance page and selects a course/session.	The system displays the correct attendance records.
Test Procedure Steps	
1	Log in as a Student or Lecturer.
2	Navigate to the Attendance page.
3	Select a course or session if required.
4	Verify that attendance records are loaded.
5	Check that the displayed status is correct, such as Present, Late, or Absent.
Post-condition	
Attendance records are displayed accurately for the selected user or course.	

Table 86: Test Case Attendance Module – MySmartStudyApp

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:149/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

4.2.35 Test Case: Certificate Module

Tested By:	Akmal Hakimi Bin Abd Rashid
Test Type	Black Box Testing – Functional Testing
Test Design Technique	Use Case Testing
Risk Number	3 (Medium - Achievement and completion feature)
Test Case Number	TC_051.1
Test Case Name	Student: Claim Certificate Successfully
Requirement Traceability	Student accesses Claim Certificate feature (Certificate Use Case)
Test Case Description	To verify that a student can claim a certificate after meeting course completion requirements.
Item(s) to be tested	
1	Certificate Eligibility Check
2	Claim Button
3	Certificate Record Generation
Pre-condition:	
Student is logged in and has completed the required course conditions.	
Specifications	
Input	Expected Output/Result
Student clicks Claim on an eligible certificate.	The system generates and stores the certificate successfully.
Test Procedure Steps	
1	Log in as a Student.
2	Navigate to the Certificates page.
3	Select an eligible certificate.
4	Click the Claim button.
5	Verify the system displays a success message.
6	Verify the certificate appears in the earned certificate list.
Post-condition	
The student certificate is generated and recorded successfully in the system.	

Table 87: Test Case Certificate Module – MySmartStudyApp

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:150/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

4.2.36 Test Case: AI Daily Study Guide Module

Tested By:	Akmal Hakimi Bin Abd Rashid
Test Type	Black Box Testing – Functional Testing
Test Design Technique	Use Case Testing
Risk Number	4 (High - Important academic workflow)
Test Case Number	TC_052.1
Test Case Name	Student: Generate AI Daily Study Guide Successfully
Requirement Traceability	Student accesses AI Daily Study Guide feature (AI Use Case)
Test Case Description	To verify that a student can generate a daily study guide using valid study input.
Item(s) to be tested	
1	Study Input Form
2	Generate Guide Button
3	AI Response Display
Pre-condition:	
Student is logged in and has access to the AI Daily Study Guide feature.	
Specifications	
Input	Expected Output/Result
Student selects a quiz, answers all questions, and clicks Submit.	The system generates and displays a personalized AI daily study guide.
Test Procedure Steps	
1	Log in as a Student.
2	Navigate to the AI Daily Study Guide page.
3	Enter valid study information.
4	Click Generate Guide.
5	Verify that the AI-generated guide is displayed.
Post-condition	
The student receives a study guide generated by the AI feature.	

Table 88: Test Case AI Daily Study GuideModule – MySmartStudyApp

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:151/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

4.2.31 Test Case: AI Exam PlannerModule

Tested By:	Mohamed Azeem bin Sulaiman
Test Type	Black Box Testing – Functional Testing
Test Design Technique	Use Case Testing
Risk Number	4 (High - Important academic workflow)
Test Case Number	TC_053.1
Test Case Name	Student: Generate AI Exam Planner Successfully
Requirement Traceability	Student accesses AI Exam Planner feature (AI Use Case)
Test Case Description	To verify that the system generates an AI exam study plan from valid exam details.
Item(s) to be tested	
1	Exam Input Form
2	Generate Plan Button
3	AI Planner Output
Pre-condition:	
Student is logged in and exam information is available.	
Specifications	
Input	Expected Output/Result
Student enters valid exam subjects and dates, then clicks Generate Plan.	The system generates and displays an AI exam planner successfully.
Test Procedure Steps	
1	Log in as a Student.
2	Navigate to the AI Exam Planner page.
3	Enter exam subject and exam date details.
4	Click Generate Plan.
5	Verify that the planner output is displayed.
Post-condition	
The AI-generated exam plan is displayed successfully.	

Table 83: Test Case AI Exam Planner Module – MySmartStudyApp

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:152/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

4.2.38 Test Case: AI Study Materials Module

Tested By:	Akmal Hakimi Bin Abd Rashid
Test Type	Black Box Testing – Functional Testing
Test Design Technique	Use Case Testing
Risk Number	4 (High - Important academic workflow)
Test Case Number	TC_054.1
Test Case Name	Student: Generate AI Study Materials Successfully
Requirement Traceability	Student accesses AI Study Materials feature (AI Use Case)
Test Case Description	To verify that a student can generate study materials such as summaries, flashcards, or practice quizzes using AI.
Item(s) to be tested	
1	Content Input/Selection
2	Material Type Selection
3	AI Output Generation
Pre-condition:	
Student is logged in and has study content or selected resources available.	
Specifications	
Input	Expected Output/Result
Student selects Summary or Flashcards and submits valid content.	The system generates the requested AI study material successfully.
Test Procedure Steps	
1	Log in as a Student.
2	Navigate to the AI Study Materials page.
3	Choose the material type such as Summary or Flashcards.
4	Provide valid study content or select a resource.
5	Click Generate.
6	Verify that the generated material is displayed correctly.
Post-condition	
The requested AI study material is generated and displayed successfully.	

Table 90: Test Case AI Study Materials Module – MySmartStudyApp

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:153/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

4.2.39 Test Case: AI Plagiarism Detection Module

Tested By:	Mohamed Azeem bin Sulaiman
Test Type	Black Box Testing – Functional Testing
Test Design Technique	Use Case Testing
Risk Number	4 (High - Important academic workflow)
Test Case Number	TC_055.1
Test Case Name	Lecturer: Run AI Plagiarism Detection Successfully
Requirement Traceability	Lecturer accesses AI Plagiarism Detection feature (AI Use Case)
Test Case Description	To verify that a lecturer can analyze a student submission using AI plagiarism detection.
Item(s) to be tested	
1	Submission Selection
2	Run Detection Button
3	Plagiarism Report Display
Pre-condition:	
Lecturer is logged in and at least one student submission is available.	
Specifications	
Input	Expected Output/Result
Lecturer selects a valid submission and runs AI plagiarism detection.	The system generates and displays a plagiarism analysis report successfully.
Test Procedure Steps	
1	Log in as a Lecturer.
2	Open the selected assignment submission page.
3	Select a student submission.
4	Click Run AI Plagiarism Detection.
5	Answer all quiz questions.
6	Wait for the report to load.
7	Verify the system displays a success message or result page.
8	Verify that the plagiarism result is displayed.
Post-condition	
The plagiarism report is generated and linked to the selected submission.	

Table 83: Test Case AI Plagiarism Detection Module – MySmartStudyApp

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:154/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

4.2.40 Test Case: AI Grading Assistant Module

Tested By:	Akmal Hakimi Bin Abd Rashid
Test Type	Black Box Testing – Functional Testing
Test Design Technique	Use Case Testing
Risk Number	4 (High - Important academic workflow)
Test Case Number	TC_056.1
Test Case Name	Lecturer: Use AI Grading Assistant Successfully
Requirement Traceability	Lecturer accesses AI Grading Assistant feature (AI Use Case)
Test Case Description	To verify that a lecturer can receive AI-generated grade suggestions and feedback recommendations for a student submission.
Item(s) to be tested	
1	Submission Analysis Trigger
2	AI Recommendation Display
3	Apply/Edit Recommendation Function
Pre-condition:	
Lecturer is logged in and a student submission is available for grading.	
Specifications	
Input	Expected Output/Result
Lecturer selects a valid submission and requests AI grading assistance.	The system generates grade suggestions and feedback successfully.
Test Procedure Steps	
1	Log in as a Lecturer.
2	Open a student submission.
3	Click Use AI Grading Assistant.
4	Wait for the system to generate grading suggestions.
5	Verify that the AI-generated mark suggestion and feedback are displayed.
6	Click the Submit button.
7	Optionally edit the recommendation and save the final result.
Post-condition	
The AI grading recommendation is displayed and can be used in the grading workflow.	

Table 83: Test Case AI Grading Assistant Module – MySmartStudyApp

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:155/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

5.0 TEST ENVIRONMENT

This section describes the required resources, configurations, and conditions necessary to execute the testing activities for the MySmartStudy Mobile and Web Application. The test environment includes hardware, software, network setup, facilities, testing tools, and any special requirements that ensure accurate and reliable test execution.

5.1 Testing Sites

5.1.1 Developer / Testing Site

Testing for the MySmartStudy application will be conducted in a controlled test environment. The environment will simulate normal usage conditions to ensure consistent performance of features such as authentication, messaging, notifications, assignments, quizzes, analytics, and data synchronization across web and mobile platforms.

Testing will be carried out using:

- Developer desktop workstations
- Android and iOS test devices (phones/tablets)
- Laptops for web-based testing
- Internet connection for Firebase Authentication, Firestore, and backend API access.
- Access to testing servers, development tools, and Git.

This environment allows testers to evaluate the application across multiple platforms and ensure seamless cross-device functionality.

5.2 Facilities Required

Physical Space

A laboratory of approximately 500 square feet, with at least 200 square feet dedicated to desk space for test devices and equipment. The area must be arranged to support collaborative testing sessions, debugging discussions, and review meetings.

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:156/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

Furniture

- Tables and chairs arranged for convenient testing activities
- Seating positioned to support communication between testers and developers
- Sufficient desk space for multiple laptops, mobile devices, and documentation

Electrical & Connectivity Needs

- Multiple power plug points near each workstation
- Stable Wi-Fi or LAN connection to enable access to Firebase services, backend API, and web/mobile testing
- Charging extensions for phones and tablets

Equipment

The following items are essential for executing test cases related to login, collaboration, Thinking Maps, assignments, notifications, and analytics:

- Desktop computers with updated browsers
- Laptops for web version testing
- Android smartphones
- iOS smartphones
- Tablets (optional but recommended)
- Headsets for team communication (if needed)

Environment Conditions

The placement of furniture and equipment should encourage smooth teamwork. Tables should be close enough to allow testers, developers, and the test lead to easily share information, observe issues together, and conduct meetings when complex test cases or defects arise.

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:157/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

This setup ensures a productive and professional environment for testing MySmartStudy's multi-platform features consistently and accurately.

5.3 Hardware and Software Preparation

This section outlines the required hardware and software components that must be prepared before executing the MySmartStudy application testing.

5.3.1 Hardware Preparation

No.	Hardware Component	Specification / Purpose
1	Desktop Computer / Laptop	Minimum Intel i5 / AMD equivalent, 8GB RAM; used for web testing and Firebase console access.
2	Android Smartphone	Android OS version 10 or higher for mobile MySmartStudy app testing.
3	iOS Smartphone	iOS 13 or higher to verify iPhone functionality and cross-platform support.
4	Tablet (Optional)	Tests UI responsiveness and map editing on larger displays.
5	High-Speed Internet Connection	Required for authentication, API communication, database access, notifications, and synchronization.
6	Power Extensions & Charging Cables	Ensure continuous device availability during test cycles.
7	Router / Wi-Fi Access Point	Provides stable connectivity for all mobile and web test devices.

Table 93: Hardware and Software Preparation - MySmartStudyApp

5.3.2 Software Preparation

No.	Software Component	Purpose / Usage
-----	--------------------	-----------------

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:158/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

1	MySmartStudy Mobile App (APK/IPA)	Main application under test for Android and iOS.
2	Web Browser (Chrome, Firefox, Edge)	Tests the MySmartStudy web dashboard for lecturer and Student functions.
3	Firebase Console	Monitors Authentication, Firestore real-time updates, and Cloud Messaging notifications.
4	Android Studio (Optional)	Installing APK, running emulators, monitoring logs, and debugging.
5	Xcode (Optional)	Used for iOS build deployment and simulator testing.
6	Postman	Used to validate API connections or backend endpoints.
7	Microsoft Excel / Google Sheets	Used to record test cases, defect logs, and result tracking.
8	Screenshot / Screen Recording Tools	Used to capture evidence of defects during testing.

Table 94: Software Preparation - MySmartStudyApp

5.4 Test Tools

This section identifies the testing tools required to support the execution of test activities for the MySmartStudy Application. These tools help testers validate authentication, database operations, notifications, UI behavior, and overall system performance. The tools listed below are essential for efficient defect detection, debugging, and documentation.

Test Sequence Table (Based on Risk Analysis)

No.	Feature	Tester
1	Register	Akmal
2	Login	Akmal
3	Create Thinking Map	Akmal
4	Real-Time Collaboration	Akmal
5	Access Class Dashboard (Lecturer)	Azeem
6	Submit Assignment	Azeem
7	Check Analytics / Learning Insights	Azeem
8	Check Study Progress	Azeem

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:159/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

9	View Recommendations	Akmal
10	Manage Classes (Lecturer)	Akmal
11	Access Resource Library	Azeem
12	Update User Profile / Settings	Akmal
13	Manage Feedback (Lecturer)	Azeem
14	Check Achievements & Badges	Azeem
15	AI Daily Study Guide	
16	AI Exam Planner	
17	AI Study Materials	
18	AI Plagiarism Detection	
19	AI Grading Assistant	

Table 95: Test Sequence Table based on risk analysis - MySmartStudyApp

5.5. Preparation and Training Required of the Test Team

When assembling the testing team, it is crucial that members receive adequate training to ensure testing is carried out accurately, consistently, and with high quality. Proper preparation enables the team to fully understand the core functions and workflows of the MySmartStudy Mobile & Web Application, allowing them to execute test conditions effectively and evaluate system behavior with confidence.

Test Planner / Controller – Azeem

- Ensure Phase 1 testing is delivered according to schedule and quality standards
- Produce high-level and detailed test conditions for all MySmartStudy modules
- Produce expected results for each designed test case
- Report progress during regular status reporting meetings
- Coordinate the review and sign-off of test conditions
- Manage individual test cycles and resolve tester queries or problems

Lead Tester – Akmal

- Identify and prepare necessary test data
- Execute assigned test conditions and record results
- Prepare software error reports with clear and complete defect details
- Administer the error measurement and tracking system
- Ensure system outages or problems are reported immediately and followed up

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:160/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

- Ensure entrance criteria are fulfilled before system testing begins
- Ensure exit criteria are fulfilled before system testing is approved for signoff

Test Designer – Azeem

- Design test components such as test cases, test scenarios, test data, test procedures, and test sequences
- Determine appropriate testing techniques, tools, and guidelines needed to implement the test plan
- Ensure test designs follow the requirements outlined in the SRS and support the overall testing strategy

6.0 TEST SCHEDULE

This section summarizes the project schedule, highlighting the academic development and milestones that associate with quality assurance.

Academic Milestone	Scope of Work / Activity	Date	Milestone
FYP Briefing	Planning & Project Feasibility Planning is the process of determining the general scope of a project and how to carry it out. Additionally, to do research on connected phrases and subjects (Note-taking habits, Thinking Maps). Next, approach a supervisor and explain the concept and its feasibility.	Start from (3/10/2025)	• Feasibility Studies • Agreed Proposal • Concept Approval • Supervisor Agreement

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:161/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

	Finally, write a proposal and get supervisor approval for the idea. For test development, this is where test plan knowledge is obtained and researched on how to write a test plan (Black Box Testing).		
Defend Proposal	Analysis & Requirement Study After proposal defence, next is a process of gathering, organizing, documenting and analyzing all the user and technical requirements (for Students and Lecturers). For test development, this phase outlines the test plan structure in terms of what to cover (e.g., Test Scope, Risk Analysis for "Recommendation Engine" and "Collaboration").	Start from (03/11/2025)	• Software Requirement Specification (SRS) • Software Test Plan (STP) • Approved Proposal
Complete Documentation	Design & Setup IT Infrastructure This activity focuses on the process of transforming user requirements into software artifact and developing technical detailed design of system architecture (e.g.,	Start from (01/12/2025)	• UML Diagram • ERD Diagram • Redefined SRS • Finalized STP

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:162/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

	Firebase Schema, Flutter Architecture). For test development, writing test plan is almost finalised to be shown to supervisor development progress.		
Progress Demo to Supervisor and Assessor	Develop & Setup Application Development activity where it emphasizes on coding, integrate testing, and implementing the component into a whole mobile app system ². For test development, test plan version 1.0 is completed and shown to supervisor³. <i>MySmartStudy Focus: Presenting the functional Map Editor and Student Dashboard prototype.</i>	Start from (16/01/2026)	• User Interface • Prototype App • Database (Firebase Setup)
Continue Development Progress	Continue App Development Still in software development. For test development, test environment preparation⁴.	Start from (17/01/2026)	• Prototype App • Database • User Interface

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:163/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

	<i>MySmartStudy Focus:</i> Integrating the Recommendation Engine logic and finalizing the Lecturer Analytics dashboard.		
Testing	Testing Acceptance Testing Updated STP for test development, this is the execution phase of planned testing for acceptance testing. Also, document the execution phase and organize it well for developer to refer later⁵. <i>MySmartStudy Focus:</i> Executing the Test Cases (TC_001 to TC_016) for Map Creation, Collaboration, and Grading features.	Start from (22/04/2026)	• Test log • Test Summary report
FYP2 Presentation	Deployment This activity process emphasizes on deployment of the system so that it can be deployed smoothly ⁶. For test development, prepare all necessary documents before end⁷. <i>MySmartStudy Focus:</i> Deploying the web portal to Firebase Hosting	Start from (02/06/2026)	• Product release

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:164/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

	and generating the final mobile APK.		
--	--------------------------------------	--	--

Table 96: Test Schedule – MySmartStudyApp

7.0 DEFINITIONS AND ABBREVIATION

7.1 Definitions

Term	Definition
Coding	Generation of source code (Dart/Flutter, PHP).
Defect	Non-conformance to requirements or functional / program specification.
Functional Testing	Testing the features and operational behaviour of a product Creating a Map, Login to ensure they correspond to its specifications.
Risk Analysis	The process of assessing identified risks to estimate their impact and probability of occurrence (likelihood).
Software Requirements Specification	A deliverable that describes all data, functional and behavioural requirements, all constraints, and all validation requirements for software.
Test Item	The individual element to be tested Recommendation Engine. There usually is one test object and many test items.
Tester	A technically skilled professional who is involved in testing of a component or system.

Table 97: Definitions – MySmartStudyApp

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:165/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

7.2 Abbreviations

Abbreviation	Meaning
ATP	Acceptance Test Plan
SRS	Software Requirements Specification
IEEE	Institute of Electrical and Electronics Engineers
ISO	International Organization for Standardization
ISTQB	International Software Testing Qualification Board
FYP	Final Year Project
IPG	Institut Pendidikan Guru

Table 98: Abbreviations – MySmartStudyApp

	ORGANISATION:	IPG KAMPUS PEREMPUAN MELAYU MELAKA	SECTION:SE
	AREA BASED ON:	MELAKA	PAGE:166/167
	DOCUMENT TITLE:	ACCEPTANCE TEST PLAN FOR MYSMARTSTUDY: DIGITAL PLATFORM FOR ENHANCING LEARNING SKILLS AMONG FIRST YEAR IPG KAMPUS PEREMPUAN MELAYU MELAKA	

